Modern Algebra II

KNU Math 415

Classnotes

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These notes are for a course taught from Fraleigh's "A First Course in Abstract Algebra" 7th edition. We refer to this book as 'the text'. These notes do not add much to the text, except sometimes a different perspective and a culling of material. We are careful to frequently reference the text for proofs and problems so that the notes can only be used with the text. You will find that many of my problems are just examples from the book.

This course takes up from Part IV (Section 18) of the text, and assumes the student has taken a course covering the first three parts of the text. It assumes that you took it from Professor Yang, with all the implications, good or bad, that that entails.

18 Rings and Fields

Recall that we gave two basic examples of groups on the set of real numbers:

$$\langle \mathbb{R}, + \rangle \qquad \text{and} \qquad \langle \mathbb{R}, \cdot \rangle.$$

Generally when we work with the reals, we use both of these groups. We use both operations, addition and multiplication.

A set with two group operations that 'work well' enough together will be a ring. A field will be the nicest of rings.

Definition 18.1. A *ring* $\langle R, +, \cdot \rangle$ is a set R with two binary operations '$+$' and '$\cdot$', referred to as *addition* and *multiplication* respectively, which satisfy three conditions:

- $\langle R, + \rangle$ is an abelian group,
- $\langle R, \cdot \rangle$ is a semigroup, that is the operation is associative:
$$a(bc) = (ab)c,$$
- and multiplication distributes over addition:
$$(a+b)c = ac + bc \qquad \text{and} \qquad c(a+b) = ca + cb.$$

Note

As we did for multiplicative groups, we write ab for $a \cdot b$.

Example 18.2. The following sets with their normal addition and multiplication operations, are rings:

$$\mathbb{Z}, \mathbb{R}, \mathbb{C}, M_n(\mathbb{R}).$$

We argued before that $\langle X, + \rangle$ is a group for each of the sets $\mathbb{Z}, \mathbb{R}, \mathbb{C}$ and $M_n(\mathbb{R})$, and that $\langle X, \cdot \rangle$ is when $X = \mathbb{R}$ or $\mathbb{C}$. So really we only have to check the associativity of $\cdot$ on $\mathbb{Z}$ and $M_n(\mathbb{R})$ and that the operations distribute. For $\mathbb{Z}$ these are well known properties. (As with the addititve properties, the actual proof of these properties requires their definition, but this gets too deep into set theory for us here, so lets just say we know them.) For $M_n(\mathbb{R})$ we these properties in linear algebra when we showed that $M_n(\mathbb{R})$ is a vector space.

As shorthand, we often say 'Let R be a ring.' instead of 'Let $\langle R, +, \cdot \rangle$ be a ring.' When we call refer to some well known set, such as $\mathbb{Z}$ as a ring, we mean with the usual addition and multiplication operations. If we want to make some other addition operation, then we will say so.

Problem 18.1. Pick some other abelian group you have seen that we write additively. Usually these have a familiar notion of multiplication. Check if this makes a ring. (Some possible groups to try: $n\mathbb{Z}$ or $\mathbb{Z}_n$ or the group F of functions from $\mathbb{Z}$ to $\mathbb{Z}$.)

As the addition operation of a ring R is a group operation, there is an identity and inverses. We refer to the identity as *the additive identity* or *the zero*, and denote it 0, and we refer to the inverse $-a$ of an element a as its *additive inverse*. We write $a - b$ for $a + (-b)$.

By definition, the additive identity 0 and the inverses satisfy the following identities.

- $0 + a = a = a + 0$
- $a - a = a + (-a) = 0 = (-a) + a$

From these properties we get some common interactions of the two operations.

Theorem 18.3. *Let R be a ring. For any $a, b \in R$ the following hold.*

i). $0a = 0 = a0$.

ii). $a(-b) = -(ab) = (-a)b$

iii). $(-a)(-b) = ab$.

Proof. See text for proof. Theorem 18.8.

$\square$

Nice! This allows us to cancel and distribute signs the way we are used to.

We have a bit of an notation issue. For additive groups, we found it useful to write $5 \cdot a$ as shorthand for

$$a + a + a + a + a.$$

This could cause confusion as '$\cdot$' is our multiplication operation. But 5 might not be in the ring, or even worse, it could be, and $5a$ might not be $a+a+a+a+a$. As a stop-gap we follow the text and write $5a$ for multiplication and $5 \cdot a$ for $a + a + a + a + a$. Usually it won't cause problems though, as usually if 5 is in our ring, we will get $5 \cdot a = 5a$.

Definition 18.4. Let R be a ring.

i). If multiplication commutes, then R is a *commutative ring.* (We don't ever say an abelian ring!)

ii). If R has a multiplicative identity, it is a *ring with unity.* The multiplicative identity is denoted 1, or occasionally 1_R.

iii). Any element with a multiplicative inverse is a *unit.* If all non-zero elements are units, then R is a *division ring.*

iv). A commutative division ring is a *field.* A non-commutative division ring is a *skew-field.*

Notationally, we use a^{-1} for the inverse of a. Writing $\frac{b}{a}$ would be unclear in general, as it could mean $a^{-1}b$ or ba^{-1}, and these are not necessarily the same. But in a commutative ring they are, so in this setting, the notation $\frac{b}{a}$, or b/a is used for $a^{-1}b$.

Now, as long as R has unity, then interpreting an element 5 as $5 = 1 + 1 + 1 + 1 + 1$, we get that

$$5 \cdot a = a + a + a + a + a = (1 + 1 + 1 + 1 + 1)a = 5a.$$

Problem 18.2. Put a check in the box of the following table if the ring in the column header is the kind of ring described in the row header. $F_{\mathbb{Z}}$ is the ring of functions from $\mathbb{Z}$ to $\mathbb{Z}$.

	$\mathbb{Z}$	$\mathbb{R}$	$M_n(\mathbb{R})$	$F_{\mathbb{Z}}$	$\mathbb{Z}_n$
commutative ring					
ring with unity					
division ring					
field					

Problem 18.3. In $\mathbb{Z}_{20}$ what are $7 \cdot 7$ and $18 + 11$?

There are a couple more basic definitions about rings that are obvious extensions of the defintions for groups. Well, I claim they are obvious. Are they? Fill them in, and then check the text to see if they are what you thought they were.

Definition 18.5. For rings $R_1, \ldots, R_n$ the direct product is the ring

$$R_1 \times R_2 \times \cdots \times R_n$$

with elements:

Addition is:

Multiplication is:

Problem 18.4. What is the 0 of $\mathbb{R} \times \mathbb{R}$? Is it a field? What is its multiplicative identity? Is $R \times R$ isomorphic to $\mathbb{C}$? Oh wait, what does isomorphism mean?!?

Definition 18.6. For rings R and R' a map $\phi : R \to R'$ is a *(ring) homomorphism* if

i). $\phi(a + b) = \phi(a) + \phi(b)$ for all $a, b \in R$, and

ii).

A (ring) homomorphism is an *(ring) isomorphism* if it is bijective.

The first condition is that ϕ is a homomphism of groups. So we know from what we proved about groups that, say,

- $\phi(0) = 0$, and
- ϕ is a bijection if and only if $\ker(\phi) = \{0\}$.

Problem 18.5. Show that the map $\phi : \mathbb{Z} \to \mathbb{Z}_n$ to the integers modulo n defined by $\phi(x) = x \mod n$ is a ring homomorphism. (Hint: How did we define this function $x \pmod n$? It'll be tricky if you don't know that.)

The kernel of this homomorphism is $n\mathbb{Z}$, which is also a ring. We will actually see that $\mathbb{Z}_n$ is the factor-ring $\mathbb{Z}_n = \mathbb{Z}/n\mathbb{Z}$, in analogue to our notion of

a factor group. (In fact this seems like such a natural place to go now, that I am rearranging the sections a bit to do this next.)

One last defintion.

Definition 18.7. A subset $R' \subset R$ is a *subring* of R if ...

In general when we define an OBJECT, a sub-OBJECT will be defined in this way.

Problems from the Text

Section 18: 5, 8, 10, 11, 23, 33, 35, 39, 44, 46, 47, 51

26 Homomorphisms and Factor rings

We just defined products and homomorphisms in analogue with out definitions for groups. It seems natural now to see a how couple of the basic results about these definitions for groups, in particular the homomorphism theorems, have ring analogues.

Problem 26.1. For a product $P := R_1 \times R_2 \times \cdots \times R_d$ of rings, show that the *projection map* $\pi_i : P \to R_i$ onto the ith component, defined by $\pi_d(r_1, r_2, \ldots, r_d) = r_i$ is a (ring) homomorphism.

The following is a collection of statements for rings, the group analogues of which are proved in Section 13.

Theorem 26.1. *Let $\phi : R \to R'$ be a ring homomorphism, and let S and S' be subrings of R and R' respectively. Let r be an element of R.*

i). $\phi(0) = 0$.

ii). $\phi(-r) = -\phi(r)$.

iii). $\phi[S]$ *is a subring of* R'.

iv). $\phi^{-1}[S']$ *is a subring of* R.

v). $\phi(1)$ *is the unit of* $\phi(R)$.

Proof. □

With these simple properties done, we move to the homomorphisms theorems of Section 14 which give a correspondence between the homomorphic images of a group, and its factor groups. This begins with defining a factor ring.

Recall that for a normal subgroup $H \leq G$ of a group G, the factor group G/H had elements $(a + H)$ for $a \in G$ and operation

$$(a + H) + (b + H).$$

We try to do the same thing with a subring $H \leq R$ of a ring R. The additive group of H is commutative, so a normal subgroup of the additive group of R. We can thus make a ring of the qoutient group R/H by defining a multiplication operation. The obvious operation is

$$(a + H)(b + H) = (ab + H),$$

as this would clearly be associative and distribute with addition. But it is not always well defined. We need a bit more.

Definition 26.2. An additive subgroup N of a ring R is an *ideal* if

$$rN \subset N \qquad \text{and} \qquad Nr \subset N$$

for all $r \in R$.

We denote an ideal by N as it is the ring analogue of the normal subgroup:

Theorem 26.3. *For a subring H of R, multiplication on the additive cosets of H is well defined by*

$$(r + H)(s + H) = rs + H$$

if and only if $rH \subset H$ and $Hr \subset H$ for all $r \in R$.

Proof. See Theorem 26.9 of text.

□

We thus get the following defintion.

Definition 26.4. For any normal subring $N \leq R$ of a ring R, the factor ring R/N has additive cosets $(r+N)$ for elements and operations $(r+N)+(s+N) = (r + s + N)$ and $(r + N)(s + N) = (rs + N)$.

As kernels of group homomorphisms were normal, we observe that kernels of ring homomorphisms are ideals.

Definition 26.5. The *kernel* of a homomorphism $\phi : R \to R'$ is the set

$$\ker(\phi) = \phi^{-1}[0];$$

exactly its kernel as a homomorphism of the additive groups.

Lemma 26.6. *The kernel of a ring homomorphism $\phi : R \to R'$ is an ideal.*

Proof. As $\{0\}$ is a subring of R', $H := \ker(\phi) = \phi^{-1}(0)$ is a subring of R. We check it is an ideal. For $r \in R$ and $h \in H$ we have that $\phi(rh) = \phi(r)\phi(h) = \phi(r)0 = 0$ and so $rh \in \ker(\phi) = H$. Thus H is an ideal by Theorem 26.3. □

Now for a homomorphism $\phi : R \to R'$ we have that the factor group $R/\ker(\phi)$ is a factor ring. With this we can state and prove the Homomorphism Theorem for Rings.

Theorem 26.7. *Let $\phi : R \to R'$ be a ring homomorphism with kernel N. The map $\mu : R/N \to \phi[R]$ defined by $\mu(r + N) = \phi(r)$ is a ring isomorphism. The map $\gamma : R \to R/N$ defined by $\gamma(r) = r + N$ is a ring homomorphism. Furthermore ϕ factors through R/N: $\phi = \mu \circ \gamma$.*

Proof. That μ is an is an isomorphism is almost done. It is an isomorphism of the additive subgroup, so it is bijective and commutes with addition. We just have to observe that it also commutes with multiplication:

$$\mu(r)\mu(s) = (r + H)(s + H) = (rs + H) = \mu(rs).$$

That γ is a homomorphism is also nearly done, we already know that it is a homomorphism of groups, so again we just have to show that it commutes with multiplication:

$$\gamma(rs) = rs + N = (r + N)(s + N) = \gamma(r)\gamma(s).$$

That ϕ factors through R/N is trivial:

$$\mu\gamma(r) = \mu(r + N) = \phi(r).$$

□

Note

The map γ is called the *(canonical) quotient map* and often denoted with a q. The map μ is called the *natural isomorphism*. The text uses Greek letters here, saving q for other things. Not a bad policy. But don't be afraid if I use a q on the chalkboard.

Problem 26.2. Show that $n\mathbb{Z}$ is an ideal of $\mathbb{Z}$. Find a ring R and a subgroup H of its additive group such that H is not an ideal.

Example 26.8. Recall that $n\mathbb{Z}$ is an ideal of $\mathbb{Z}$. By the Homomorphism Theorem we conclude that $\gamma_n : \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z} : z \mapsto z \pmod{n}$ is a homomorphism, and so

$$\mu_n : \mathbb{Z}/n\mathbb{Z} \to \mathbb{Z}_n : z + n\mathbb{Z} \mapsto z \pmod{n}$$

is an isomorphism. The ideal $n\mathbb{Z}$ is the kernel of their composition ϕ_n, our usual 'mod n' map

$$\phi_n : \mathbb{Z} \to \mathbb{Z}_n : z \mapsto z \pmod{n}.$$

Problems from the Text

Section 26: 1, 3, 4, 5-7, 9, 10, 13, 17, 21, 27, 29, 34-36

19 Integral Domains

Remember when you first learned about groups, and you always assumed that $a \cdot b = b \cdot a$. You didn't believe there were non-abelian groups. Then Professor Yang reminded you that $M_3(\mathbb{R})$ is a group under multiplication, and you had to admit, 'Oh yeah.' Perhaps you cursed him under your breath, and whispered that that was a dirty trick. I agree. It was. And I have some other dirty tricks for you.

i). We can't assume that $ab = 0$ imples that $a = 0$ or $b = 0$.

ii). You can't alway cancel in rings: $ab = ac$ does not always imply that $b = c$ when $a \neq 0$.

Problem 19.1. Give examples in $\mathbb{Z}_n$, for some n, showing that the statements i) and ii) above are true. Give examples in $M_2(\mathbb{Z})$.

Definition 19.1. A non-zero element a of a ring R is a *zero divisor* if there is a non-zero element b with $ab = 0$ or $ba = 0$.

Problem 19.2. Find a ring R with elements a and b such that $ab = 0$ but $ba \neq 0$. Challenge: Find a ring R and an element $a \in R$ such that $ab = 0$ for some b but for which there is no b' such that $b'a = 0$.

Theorem 19.2. *An element of $\mathbb{Z}_n$ is a zero divisor if and only if it is not relatively prime to n.*

Proof. See text. This is Theorem 19.3. □

Corollary 19.3. *If p is prime, then $\mathbb{Z}_p$ has no zero divisors.*

Theorem 19.4. *If a ring R has no zero divisors if and only if it has* cancellation*:*

i). $ab = ac$ *and* $a \neq 0$ *imply that* $b = c$

ii). $ba = ca$ *and* $a \neq 0$ *imply that* $b = c$

Proof. Assume that R has no zero divisors, and that $ab = ac$ for some $a \neq 0$. Then

$$0 = (ab - ac) = a(b - c),$$

and so $b - c = 0$ meaning that $b = c$.

Assume on the other had that R has cancellation, and towards contradiction that a is a zero divisor. Then for some non-zero b we have

$$ab = 0 = a0.$$

But cancelling gives $b = 0$, a contradiction. □

Definition 19.5. A commutative ring D with unity and with no zero divisors is an *integral domain*.

We call it this, because it acts very much like the integers.

Problem 19.3. Show that $\mathbb{Z}_n$ is not an integral domain if n is not prime. Is $\mathbb{Z} \times \mathbb{Z}$ an integral domain?

We use the word *subdomain* instead of sub-(integral domain).

Theorem 19.6. *Every field is an integral domain. Every finite integral domain is a field.*

Proof. See text. Theorems 19.9 and 19.11.

□

Corollary 19.7. *If p is a prime, then $\mathbb{Z}_p$ is a field.*

Definition 19.8. The *characteristic* of a ring R is the minimum positive integer n, if it exists, such that $n \cdot r = 0$ for all r in R. If such n does not exist, then the characteristic of R is 0.

Problem 19.4. What is the characteristic of $\mathbb{Z}, \mathbb{R}, \mathbb{Z}_8, \mathbb{Z}_5 \times \mathbb{Z}_{20}$?

Theorem 19.9. *In a ring R with unity, the characteristic of R is n if n is the minimum positive integer such that $n \cdot 1 = 0$. If such n does not exist, then by definition the characteristic is 0.*

Proof. Let $n > 0$ be the min positive integer such that $n \cdot 1 = 0$. Then for any r

$$n \cdot r =$$

□

Problems from the Text

Section 19: 5, 7, 14, 15, 16, 17, 18, 21, 22, 23, 27, 30

20 Fermat's and Euler's Theorems

You might see a different proof of this in a number theory course, but we have built up some nice ideas, so we can use them for a quick proof.

Theorem 20.1 (Fermat's Little Theorem)**.** *For any prime p and any integer a not divisible by p we have $a^{p-1} \equiv_p 1$.*

Proof. First observe that the subset of non-zero elements F^* of a field F is a multiplicative subgroup: as F has no 0 divisors, F^* is closed under multiplication; it contains 1; and it has inverse as a non-zero element cannot have 0 as an inverse.

Applying this to $\mathbb{Z}_p$ we see that the set $\mathbb{Z}_p^* = \{1, 2, \ldots, p-1\}$ form a group of order $p-1$ under multiplication, so for any z in $\mathbb{Z}_p^*$ we have $z^{p-1} = 1$.

Now, recall that when $\psi : \mathbb{Z} \to \mathbb{Z}_p$ was the map $\psi(a) = a \mod p$, the kernel was $\ker(\psi) = p\mathbb{Z}$. For an integer a not divisible by p, a is not in this kernel, so $\psi(a)$ is in to $\mathbb{Z}_p^*$. Thus, as needed,

$$a^{p-1} \mod p = \psi(a^{p-1}) = \psi(a)^{p-1} = 1.$$

□

This gives a immediate corollary that is useful in computing modulo p.

Corollary 20.2. *For any prime p and integer a, we have $a^p \equiv_p a$.*

Proof. For a not divisible by p this is immediate from the theorem by multiplying both sides of the equation by a. If a is divisible by p, then both sides are equivalent to 0 mod p, so the result is trivial. □

Problem 20.1. Compute $5^{108} \pmod 7$.

Now, what about $\mathbb{Z}_n$ when n is not prime. We can try to do the same thing. Euler did this. Let G_n be the set of elements of $\mathbb{Z}_n^*$ that are not zero divisors. Let $\phi(1) = 1$ and for integer $n \geq n$ let $\phi(n) = |G_n|$. This is known as the *Euler phi function* or the *Euler totient function.*

Problem 20.2. When p is prime then what is $\phi(p)$? What is $\phi(10)$?

Theorem 20.3 (Euler's Theorem)**.** *Let $n \geq 2$ be an integer, and a be relatively prime to n, then $a^{\phi(n)} \equiv_n 1$.*

Proof. The proof follows that of Fermat's Little Theorem, but some of the steps take a bit more work.

First we show the following.

Claim 20.4. *G_n is a multiplicative subgroup of $\mathbb{Z}_n$*

Proof. Essentially the same proof tells us that it is closed under multiplication: (fill this in)

We know the identity 1 of $\mathbb{Z}_n$ is in G_n, but we do not know it is a field so our proof from above will not be enough to show we have inverse.

Let a be in G_n. It is not a zero divisor of $\mathbb{Z}_n$ so where $G_n = \{a_1, a_2, \ldots, a_m\}$ we have that the elements

$$aa_1, aa_2, \ldots, aa_m$$

are distinct: if $aa_i = aa_j$, then $0 = aa_i - aa_j = a(a_i - a_j)$, and as a is not a zero divisor, this means that $i = j$. So $G_n = \{aa_1, \ldots, aa_m\}$ meaning that some $1 = aa_i$ for some i. So the inverse a_i of a is in G_n. $\diamond$

Now that we know that G_n is a group, we get immediately, for any $z \in G_n$, that $z^{|G_n|} = z^{\phi(n)} = 1$.

(This feels like we are done, but the statement isn't about elements in $\mathbb{Z}_n$, it is about elements in $\mathbb{Z}$.)

To finish off, if we can show that for a relatively prime to n the image of a under the quotient map $\gamma : \mathbb{Z} \to \mathbb{Z}/n : a \mapsto a \mod n$ is in G_n then we are done because

$$a^{\gamma(n)} \mod n = \gamma(a^{\phi(n)}) = \gamma(a)^{\phi(n)} = 1.$$

So towards contradiction, let a be prime to n but assume that $\gamma(a)$ is a zero-divisor in Z_n. (It cannot be 0 as a is not divisible by n.) The for some $b \in \mathbb{Z}_n^*$ we have $\gamma(a)b = 0$. But $\gamma(b) = b$ so $\gamma(ab) = 0$ meaning that $n|ab$ in $\mathbb{Z}$. So as a is prime to n, $n|b$. But this contradicts the fact that $b \in \mathbb{Z}_n^*$ is not a zero-divisor.

$\square$

Euler's theorem has a lot of nice applications in number theory, and cryptograpy. Some are given in the text, but we will skip most of these.

Problems from the Text

Section 20: 1,4,8,9,23,25,26

Note

The notation $Q(D)$ is not in the text. It is not standard, but is convenient. The 'field of quotients' is also often called the 'field of fractions'.

21 The Field of Quotients of an Integral Domain

The integral domain $\mathbb{Z}$ is not a field– it does not have multiplicative inverses. But we introduce these, and so make the field $\mathbb{Q}$. In the same way, we can make a field of quotients $Q(D)$ from any integral domain D.

Our first step is to define the elements of $Q(D)$. Throughout the section, keep the example of $D = \mathbb{Z}$ and $Q(D) = \mathbb{Q}$ in mind.

For an integral domain D let

$$S(D) = \{\frac{a}{b} \mid a, b \in D, b \neq 0\}.$$

Define an equivalence relation $\sim$ on $S(D)$ by letting $\frac{a}{b} \sim \frac{c}{d}$ if $ad = bc$ in D. Let the elements of $Q(D)$ be the classes of the partition $S(D)/\sim$. We represent the class of $\frac{a}{b} = a/b$ by $[a/b]$.

Problem 21.1. Show that $\sim$ is indeed an equivalence relation.

Definition 21.1. For an integral domain D, the *quotient field* $Q(D)$ is the set defined above with the operations

$$[a/b]\,[c/d] = [ad + bc/bd] \qquad \text{and} \qquad [a/b]\,[c/d] = [ac/bd]\,.$$

Bad form! We made a definition without checking that it was well defined. Let's do that now.

Lemma 21.2. *The operations on $Q(D)$ from Defintion 21.1 are well defined.*

Proof. So that the elements $[ad + bc/bd]$ and $[ac/bd]$ are actually in the set $Q(D)$ we need that $bd \neq 0$, but this is okay, because we had $b \neq 0$ and $d \neq 0$. So we get this from the fact that D is an integral domain. (And this is where we really need this!)

Now suppose that $\frac{a'}{b'} \in [a/b]$ and $\frac{c'}{d'} \in [c/d]$. We must show that

$$\frac{a'd' + b'c'}{b'd'} \in [(ad + bc)/bd] \qquad \text{and} \qquad \frac{a'c'}{b'd'} \in [ac/bd]\,.$$

That is: we have that $a'b = b'a$ and $c'd = d'c$, and we have to check that

$$(a'd' + b'c')bd = b'd'(ad + bc) \qquad \text{and} \qquad a'c'bd = b'd'ac.$$

This is not so hard– remember that D is a commutative ring.

□

Now, we have called $Q(D)$ the quotient field of D, so we should check that it is, indeed, a field.

Lemma 21.3. *For an integral domain D, the set $Q(D)$ with operations from Definition 21.1 is a field.*

Proof. There are a lot of things to check here, see P. 193 of the text. We don't even have the usual head start of knowing that the addition operation is a group operation. Check this first. What is the additive identity? What is $-[a/b]$? Use $\mathbb{Q}$ as a guide, and then check that it works. Now check that the multiplication operation is associative, commutative, and distributes over addition. What is the one? What is $[a/b]^{-1}$? □

Okay. Our field $Q(D)$ of quotients of D is defined, but there are a couple of properties that we want it to have. It should, contain D as a sub-domain, and live inside any field that contains D.

Lemma 21.4. *Let D be an integral domain and $Q(D)$ be its ring of quotients. The map $i : D \to Q(D) : a \mapsto [a/1]$ is an injective homomorphism.*

Proof. To see that it is a homomorphism, we check

$$i(a) + i(b) = [a/1] + [b/1] = [(a1 + 1b)/(1 \cdot 1)] = [(a + b)/1] = i(a + b),$$

and

$$i(a)i(b) = [a/1]\,[b/1] = [ab/1] = i(ab).$$

To see that it is injective, assume that $i(a) = i(b)$, so $[a/1] = [b/1]$. Using this for the middle inequality in the following, we get

$$a = a \cdot 1 = b \cdot 1 = b,$$

which shows that i is injective. □

Now we show that $Q(D)$ injects into any field containing D.

Theorem 21.5. *Let D be an integral domain, and L be a field containing D. There is an injection $\phi : Q(D) \to L$ such that $\phi(a) = a$ for all $a \in D \subset Q(D)$.*

Proof. Set $\phi_D : D \to L : a \mapsto a$ and extend it to a map $\phi : Q(D) \to L$ multiplicatively: for any $[a/b]$ in $Q(D)$ let $\phi([a/b]) = \phi_D(a)/\phi_D(b) = a/b$. We must show that ϕ is (well defined and) an injective homomorphism.

Certainly for any $[a/b]$ in $Q(D)$ we have $b \neq 0$ and so $\phi_D(b) \neq 0$, showing that ϕ maps to L. If $[a'/b'] = [a/b]$ in $Q(D)$ then $a'b = b'a$ in $Q(D)$, so in D. To see that ϕ is well defined we must show that $\phi([a'/b']) = \phi([a/b])$ in L, which comes down to showing that that $a'/b' = a/b$ in L, or equivalently that $a'b = ab'$ in L. But we have this for free:

$$a'b = \phi_D(a'b) = \phi_D(b'a) = b'a,$$

so ϕ is well defined.

That it is a homomorphism translates transparently from the fact that operations in $Q(D)$ and L are basically the same:

$$\phi(a/b+c/d) = \phi((ad+cb)/cd) = (ad+cb)/cd = (a/b)+(c/d) = \phi(a/b)+\phi(c/d),$$

and

$$\phi((a/b)(c/d)) = \phi(ac/bd) = ac/bd = (a/b)(c/d) = \phi(a/b)\phi(c/d).$$

To see that it is an injection, assume that $\phi(a/b) = \phi(a'/b')$. Then $(a/b) = (a'/b')$ and so $ab' = ba'$ in L. But then $\phi_D(ab') = \phi_D(ba')$, and as ϕ_D was an isomorphism, $ab' = ba'$. □

Again, this says that if D is a subdomain of some field L, then we can identify $Q(D)$ with some subfield of L, so $Q(D)$ is, upto isomorphism, the unique field of quotients of D.

Problems from the Text

Section 21: 1,4,5,6-11

22 Rings of Polynomials

In this section we define a new ring $R[x]$ of polynomials from a ring R by adding a new element x, called an indeterminate.The elements of $R[x]$ look like the polynomial functions that we are familiar with, but we try not to look at them like this. We think of them as algebraic objects. We won't really be interested in what their graphs look like; we will be more interested in finding their zeros, and factoring them.

Definition 22.1. Let R be a ring. A *polynomial $f(x)$ with coefficients in R* is a formal sum

$$\sum_{i=0}^{\infty} a_i x^i = a_0 + a_1 x + a_2 x^2 + \dots$$

with a_i in R for all R and $a_i \neq 0$ for only finintely many i. The element a_i is the ith *coefficient* of $f(x)$ and the degree of $f(x)$ is the largest i such that $a_i \neq 0$. If $f(x) = 0$ then the degree is undefined.

Generally do not write the summands with a co-efficient of 0 and so we usually denote a polynomial by $f(x) = \sum_{i=0}^{n} a_i x^i$ where a_i is assumed to be non-zero. The value n is the *degree* of $f(x)$. If the degree is 0, $f(x)$ is a *constant*.

To make a ring out of the polynomials, we need some operations. These are the operations you are used to for polynomials.

Definition 22.2. *Polynomial addition and multiplication* are the operations defined for polynomials $f(x) = \sum_{i=0}^{\infty} a_i x^i$ and $g(x) = \sum_{i=0}^{\infty} b_i x^i$ by

$$f(x) + g(x) = \sum_{i=0}^{\infty} (a_i + b_i) x^i \qquad \text{and} \qquad f(x)g(x) = \sum_{i=0}^{\infty} \left(\sum_{j=0}^{i} a_j b_{i-j} \right) x^i.$$

Notice that our definition "collects the 'x's to the right". This make the indeterminate commute with everything by definition, and so if R is commutive then so is $R[x]$. Oh, to say this we should first say that $R[x]$ is a ring...

Theorem 22.3. *For a ring R, the set $R[x]$ of polynomials with co-efficients in R is a ring under polynomial addition and multiplicaton.*

Proof. That addition is a group operation is pretty straight-forword. The verification that multiplication is associative and that the operations distribute is not hard, but takes some patience and attention to indices. See the text, Theorem 22.2. □

Problem 22.1. What is the zero of $R[x]$? If R has unity, then so does $R[x]$; what is it? If R is a field, is $R[x]$?

A couple more definitions for later:

- As $R[x]$ is a ring, we can treat it as the constant ring for a ring $(R[x])[y]$ of polynomials in the indeterminate y. We write this as $R[x, y]$. Generalising, $R[x_1, x_2, \dots, x_n]$ is the *ring of polynomials in n indeterminates over R*.
- If $F$is a field, then $F[x]$ is an integral domain, and so is $F[x_1, \dots x_n]$. The field $F(x_1, \dots, x_n)$ of quotients of $F[x_1, \dots x_n]$ is the *field of rational functions in n indeterminates over F*.

We look at a polynomial $f(x)$ not as a function, but as an element of a ring $R[x]$. So what is $f(\alpha)$ for some value α. Exactly what you think it is, but we are going to define it what seems like a round about way.

Theorem 22.4. *Let F be a subfield of a field E, and let α be in E. The map $\phi_\alpha : F[x] \to E$ defined by*

$$\phi_\alpha(\sum_{i=0}^{\infty} a_i x^i) = \sum_{i=0}^{\infty} a_i \alpha^i$$

is a homomorphism that is the identity map on F.

Proof. The map clearly maps into E, and is the identity on constants in F. That it is a homomorphism is again transparent:

$$\begin{aligned} \phi_\alpha(\sum a_i x^i + \sum b_i x^i) &= \phi_\alpha(\sum (a_i + b_i) x^i) = \sum (a_i + b_i)\alpha^i \\ &= \sum a_i \alpha^i + \sum b_i \alpha^i = \phi_\alpha(\sum a_i x^i) + \phi_\alpha(\sum b_i x^i) \end{aligned}$$

and ... (fill this in.)

□

Now as expected, we define $f(\alpha)$ as $f(\alpha) = \phi_\alpha(f(x))$, we call $\alpha \in E$ a *zero* of $f(x)$ if $f(\alpha) = 0$, that is, if $f(x)$ is in the kernel of ϕ_α. From this it follows that the set of polynomials for which α is a zero is an ideal. Sure we could prove this useful fact directly without much work, but this is an example of why we have defined evaluation through the evaluation homomorphism.

A goal in the upcoming sections is address the problem of deciding if a polynomial $f(x)$ in some polynomial ring $F[x]$ has zeros in some given ring E containing F. We know, for example that $f(x) = x^2 - 2$ in $\mathbb{Q}[x]$ has no zeros in $\mathbb{Q}$, but does have them in $\mathbb{R}$. We call them $\pm\sqrt{2}$. Do you remember the proof that $\sqrt{2}$ is not in $\mathbb{Q}$? Probably, but it does not hurt to see it again.

Theorem 22.5. *The polynomial $f(x) = x^2 - 2$ has no zeros in $\mathbb{Q}$.*

Proof. Assume it has some zero $a/b \in \mathbb{Q}$. We may assume that a and b are relatively prime. Plugging into $f(x)$ we get $2 = (a/b)^2 = a^2/b^2$, so $a^2 = 2b^2$. But then 2, being prime, divides a so 2^2 divides a^2. But then 2 divides b, contradicting the fact that a and b are relatively prime. □

As $f(x) = x^2 - 2$ has no zeros in $\mathbb{Q}$, it certainly has no zeros in $\mathbb{Z}$.

Problem 22.2. Does $f(x) = x^2 - 2$ have zeros in $\mathbb{Z}_2[x]$ or $\mathbb{Z}_3[x]$?

Problems from the Text

Section 22: 3, 9, 15, 18-19, 21, 23, 24, 25, 27, 29

23 Factorisation of polynomials of a field

In this section we look at several results about polynomials over indegral domains that you are probably familiar with from high school. Using a polynomial version of the division algorithm, we show that $f(x)$ has zero a if and only if $(x - a)$ divides $f(x)$, and use this to show that a degree n polynomial has at most n zeros.

We then look at divisibilitly conditions like the Eisenstien critireum.

Finally, we (mostly) prove unique factorisation.

All of of this starts with the division algorithm.

Theorem 23.1 (The polynomial division algorithm)**.** *For a field F, let $f(x) = \sum_{i=0}^{n} a_i x^i$ and $g(x) = \sum_{i=0}^{m} b_i x^i$ be polynomials in $F[x]$ with $g(x)$ non-constant. There are unique polynomials $q(x)$ and $r(x)$ in $F[x]$ with $r(x) = 0$ or $\deg(r) < \deg(g)$ such that*

$$f(x) = q(x)g(x) + r(x).$$

Proof. The set $S = \{f(x) - g(x)q(x) \mid q(x) \in F[x]\}$ is non-empty. If it contains 0 then and we are done. So we may assume that it does not, and we can take

an element $r(x) = \sum_{i=1}^{d} c_i x^i$ of minimum degree. We have that

$$f(x) = g(x)q(x) + r(x)$$

for some $q(x)$ in $F[x]$. We show that $\deg(r) < \deg(q)$. Assume that it isn't, so $d \geq m$. Then $g(x) \cdot \frac{c_d x^d}{b_m x^m}$ has the same leading term as $r(x)$ and so $r'(x) := r(x) - g(x) \cdot \frac{c_d}{b_m}(x^{d-m})$ has degree at most $d - 1$. But then

$$f(x) = g(x)\left(q(x) - \frac{c_d}{b_m}x^{d-m}\right) + r'(x)$$

contradicts our choice of $r(x)$.

Problem 23.1. We prove the existence of $q(x)$ and $r(x)$. Show that such $r(x)$ is unique. (Assume there is another such $r'(x)$. Use a similar proof to above to show that there is another one $r''(x)$ with degree smaller than $r(x)$.) $\square$

We are now going to prove a lot of the properties of polynomials that you learned in high-school. But we will be proving them over general fields, so we have to be careful. We will skip some of the proofs. There are a lot.

Problem 23.2. If you forget how to do it from high school, use polynomial long division to divide $f(x) = x^4 - 3x^3 + 2x^2 + 4x - 1$ by $g(x) = x^2 - 2x + 3$ in $\mathbb{Z}_5[x]$. This is example 23.2 of the text. Make sure your co-efficients stay in $\mathbb{Z}_5$.

Corollary 23.2 (The Factor Theorem)**.** *An element a of a field F is a zero of $f(x) \in F[x]$ if and only if $(x - a)$ is a factor of $f(x)$ in $F[x]$.*

Proof. Try to prove this on your own before looking at the proof in the book. 'If' should be easy. For the 'only if' part, here is a hint: this is a corollary of the division algorithm; so try to use it. (Divide $f(x)$ by $(x - a)$.) □

This helps us factor things. Especially in a ring like $\mathbb{Z}[5]$. You can evaluate at each element an see if it is a zero. If it is, divide it out using long division. Then repeat. See example 23.4 of the text.

Corollary 23.3 (The Polynomial Roots Theorem)**.** *A polynomial of degree n in $F[x]$ has at most n zeros in F.*

Proof. Use an induction, and use that F is a field, so has no zero divisors. □

This one is a little trickier, but also, you maybe haven't seen it in high-school

Corollary 23.4. *If G is a finite subgroup of the multiplicative group $\langle F^*, \times\rangle$ of a field F, then G is cyclic. In particular, the multiplicative group of any finite field it cyclic.*

Proof. Let G be a finite subgroup of $\langle F^*, \times\rangle$. As F is a field G is abelian, so being finite, we saw in Section 13 that

$$G = \sum_{i=1}^{r} \mathbb{Z}_{d_i}$$

is a product of cyclic groups. Recall that the product is cyclic if and only if the d_i are powers of distinct primes if and only if $m := \mathrm{lcm}(d_1, \ldots, d_r)$ is equal to $\prod d_i$. We have easily that $m \leq \prod d_i$. On the other hand, as every element of G has order at most m, every element of G is a zero of $x^m - 1$. But this can have at most m zeros, so $\prod d_i \leq m$. □

Divisibility Conditions and irreducibility

Definition 23.5. A polynomial $f(x)$ is *irreducible over F, or irreducible in $F[x]$,* if it is non-constant and cannot be expressed as the product of two polynomials of lower degree. A non-constant polynomial that is not irreducible is *reducible.*

So $f(x) = x^2 + 1$ is irreducible over $\mathbb{R}$, but reducible over $\mathbb{C}$.

Problem 23.3. By evaluating it for all values in $\mathbb{Z}_5$ show that $f(x) = x^3+3x+2$ is irreducible in $\mathbb{Z}_5[X]$. Would this work for $g(x) = x^8 + 3x + 2$?

Theorem 23.6. *Let $f(x) \in F[x]$ have degree 2 or 3, then $f(x)$ is reducible if and only if it has a zero in F.*

Proof. What is the main idea of the proof? Here is a hint: why does the proof only work for polynomials of degree 2 and 3? See the last problem. See the text. □

Theorem 23.7. *If $f(x)$ in $\mathbb{Z}[x]$ is irreducible in $\mathbb{Z}[x]$ then it is irreducible in $\mathbb{Q}[x]$.*

Proof. The text begs off this one as being messy, so we will too. If that bugs you though, good on you. Maybe try to prove it first for $f(x)$ having degree 2. □

The following uses the above theorem and Corollary 23.2.

Corollary 23.8. *If $f(x) = \sum_{i=0}^{d} a_i x^i$ in $\mathbb{Z}[x]$ has a zero in $\mathbb{Q}$ then it has a zero in $\mathbb{Z}$ that divides a_0.*

Proof. Assume that $f(x)$ has a zero a/b in $\mathbb{Q}$. By Corollary 23.2, $f(x)$ has a factor $(x - a/b)$ in $\mathbb{Q}[x]$. If this is not a factor in $\mathbb{Z}[x]$ then it is a factor in $\mathbb{Q}[x]$ of some irreductible factor $g(x)$ of $f(x)$ in $\mathbb{Z}[x]$. But the above theorem says that $g(x)$ cannot be irreducible in $\mathbb{Z}[x]$. So $(x - a/b)$ is in $\mathbb{Z}[x]$ and $b = 1$. As $(x - a)$ divides $f(x)$, which has constant term a_0, a divides a_0. □

Problem 23.4. Use Theorem 23.7 and the corollary to show that $f(x) = x^4 - 2x^2 + 8x + 1$ is irreducible over $\mathbb{Q}$. Use the corollary to show that $x^2 - 2$ is irreducible over $\mathbb{Q}$.

Theorem 23.9 (Eisenstein's Criterion)**.** *Let $f(x) = \sum_{i=0}^{n} a_i x^i$ be in $\mathbb{Z}[x]$ and $p \in \mathbb{Z}$ be prime. If $p \mid a_i$ for all i but n, and $p^2 \nmid a_0$. Then $f(x)$ is irreducible over $\mathbb{Z}$ and so over $\mathbb{Q}$.*

Proof. See text for proof. □

Here is an application of this that I hope we use later.

We know that $x^p - 1$ has 1 as a zero, and we can easily show the following:

$$\Phi_p(x) := \frac{x^p - 1}{x - 1} = x^{p-1} + x^{p-2} + \cdots + x + 1$$

This $\Phi_p(x)$ is called the *cyclotomic polynomial* of order p.

Corollary 23.10. *The cyclotomic polynomial $\Phi_p(x)$ is irreducible over $\mathbb{Q}$.*

Proof. The proof in the text uses a neat trick. Using the binomial theorem to compute $\Phi_p(x+1)$ you get a polynomial to which you can apply the Eisenstein Criterion to show that it is not reducible. If $\Phi_p(x)$ were reducible, then $\Phi_p(x+1)$ would be too, so this gives the result. □

Unique factorisation in the polynomial ring

Definition 23.11. A polynomial $f(x) = \sum_{i=0}^{n} a_i x^i$ is *monic* if its leading coefficient a_n is 1. Constant elements of $F[x]$ are often called *units*, as they are the invertible elements of the ring.

It $F[x]$ every polynomial $f(x)$ can clearly be written as a product $a \cdot m(x)$ of a where a is a unit and $m(x)$ is a monic polynomial.

Definition 23.12. A polynomial $g(x)$ in $F[X]$ *divides* a polynomial $f(x)$ in $F[x]$, written $g(x) \mid f(x)$, if there is a $q(x)$ in $F[x]$ such that $f(x) = g(x)q(x)$

Theorem 23.13. *Let $p(x)$ be an irreducible polynomial in $F[x]$. If $p(x)$ divides $r(x)s(x)$ for polynomials r and s in $F[X]$, then $p(x)$ divides either $r(x)$ or $s(x)$.*

We put the proof of this off until the next section. It might seem easy, but the 'easy' proof of it usually uses the unique factorisation of polynomials, and this is what we are going to prove next, as a consequence of this theorem. Or it looks in a bigger field in which every polynomial factors into linear factors, and then uses the correspondence between factors and zeros. But we don't know how to do this yet!

Theorem 23.14. *For a field F, every non-constant monic polynomial $f(x) \in F[x]$ can be factored into a unique (up to order) product of irreducible monic polynomials.*

Proof. The theorem is clearly true for polynomials of degree 1. Assume we have shown it for polynomials of degree less than $n > 1$, and let $f(x)$ be a monic polynomial of degree n. If $f(x)$ is irreducible, then we are done, so assume that $f(x)$ factors as $f(x) = g(x)q(x)$ into monic polynomials g and q of degree less

than n. Each of these factors uniquely into irreducible monics, and $f(x)$ is the product of these two factorisations of irreducible monics.

Suppose there is are two such factorisations

$$p_1(x)p_2(x)\dots p_m(x) = f(x) = q_1(x)q_2(x)\dots q_n(x).$$

Then $p_1(x) \mid q_1(x)q_2(x)\dots q_n(x)$ and with Theorem 23.13 we can show that $p_1(x) \mid q_i(x)$ for some i. But $q_i(x)$ is irreducible, so $p_1(x) = q_i(x)$. But then applying the induction hypothesis to $f(x)/p_1(x)$ we get that the other factors coincide as well. □

Problems from the Text

Section 23: 1, 6, 13, 15, 23-24, 25, 26, 28, 32-33

27 Ideals and factor rings

Our main goal in this section is to prove Theorem 23.13, which says that if irreducible polynomial f divides $g \cdot h$, then it divides g or h.

Recall that an additive subgroup N of a ring R was an ideal if it multiplicatively absorbs: $aN, Na \subset N$ for all $a \in R$. It was the analogue of a normal subgroup of a group in that it was exactly the definition of a subgroups that could quotient N by to define a ring homomorphism to a factor ring R/N. In this section we restrict to commutative rings R.

The *principle ideal of R generated by an element $a \in R$* is the set $\langle a \rangle = \{ar \mid r \in R\}$. For a polynomial $f(x)$ in a ring $F[x]$, the principle ideal $\langle f \rangle$ is just the set of polynomials that it divides.

Problem 27.1. Show that for a commutative ring R with unity that the set $\{ra \mid a \in R\}$ is an ideal.

An ideal N of R is

- *proper* if N is not R, *non-trivial* if N is not 0,
- *maximal* if it is proper, non-trivial, and is contained no larger proper ideal of R; and
- *prime* if $a \cdot b \in N$ implies $a \in N$ or $b \in N$.

Problem 27.2. Show the following about the ring $\mathbb{Z}$.

i). All ideals are principle.

ii). The principle ideal $\langle a \rangle = a\mathbb{Z}$ is maximal if and only if a is prime.

iii). The principle ideal $\langle a \rangle$ is prime if and only if a is prime.

iv). An ideal N is maximal if and only if $\mathbb{Z}/N$ is a field.

Problem 27.3. Find an ideal of $\mathbb{Z} \times \mathbb{Z}$ that is not principle. Show that $\mathbb{Z} \times \{0\}$ is a prime ideal of $\mathbb{Z} \times \mathbb{Z}$. Is it maximal? Is $p\mathbb{Z} \times p\mathbb{Z}$ a prime ideal of $\mathbb{Z} \times \mathbb{Z}$? Find a maximal ideal of $\mathbb{Z} \times \mathbb{Z}$. Is it prime?

Proving Theorem 23.13 comes down to showing, for irreducible $f(x)$, that $\langle f \rangle$ is maximal, and so that $\langle f \rangle$ is prime. We do this is a somewhat round-about way breaking it down into some more general statements that will have other uses.

That fields have no proper non-trivial ideas is simple. Indeed we have the following.

Fact 27.1. *If R is a ring with unity, and N is an ideal of R containing a unit, then $N = R$. So a field has no proper non-trivial ideals.*

Proof. Assume R has an ideal N containing 1, then $r = r \cdot 1$ is in N for all $r \in R$, and so $N = R$ is not proper. If a field F has a non-trivial ideal N containing some element a, then $1 = a^1 a$ is in N, and so by the above statement $N = F$. $\square$

Theorem 27.2. *Let R be a commutative ring with unity.*

i). An ideal M is maximal if and only if R/M is a field.

ii). An ideal N is prime if and only if R/N is an integral domain.

Proof. Recall that the quotient map $\gamma : R \to R/M$ and its inverse both take ideals to ideals.

i). On the one hand, assume that M is maximal. As R/M is commutative and has a unit $1 + M$, it is a field unless it contains some $(a + M)$ without inverse. Assume it does; the primitive ideal $\langle (a + M) \rangle$ does not contain the unit $(1 + M)$, so its preimage $\gamma^{-1}(\langle (a + M) \rangle)$ is an ideal of R containing M but not containing 1, so is a proper ideal containing M. This contradicts the fact that M is maximal.

On the other hand, if M is not maximal we have a proper ideal N of R that properly contains M. As it properly contains M, $\gamma(N)$ is not 0. If $\gamma(N) = R/M$, then for every $r \in R$ we have some $n \in N$ such that $(r + M) = (n + M)$, and so $r = n + m$ for some m in M. But $n + m \subset N$ so this would mean that $r \in N$. As this is for all r, we get that $N = R$, contradicting the fact that N is proper.

ii). As $N = (0 + N)$ is the zero of R/N a zero divisor is some element $(a + N)$ such that there is another element $(b + N)$ such that

$$(ab + N) = (a + N)(b + N) = (0 + N);$$

so R/N has a zero divisor unless $ab \in N$ implies that $a \in N$ or $b \in N$.

□

As every field is an integral domain, we get:

Corollary 27.3. *Every maximal ideal of a commutative ring is prime.*

Excursion to prime fields

Now we will look at a couple of nice results we can say with the preceeding results.

Problem 27.4. Prove the following theorem and its two corollaries. They are, of course, in the text, but try on your own first. The theorem is just a standard checking of properties– nothing clever. The corollaries are short, just patching together several of the results we have proved over the last couple of sections.

Theorem 27.4. *Let R be a commutative ring with unity. The map*

$$\phi\colon \mathbb{Z} \to R : n \mapsto n \cdot 1$$

is a ring homomorphism.

The kernal of ϕ is an ideal of $\mathbb{Z}$ which must be of the form $s\mathbb{Z}$ for some integer s and so by the homomorphism theorem the image of ϕ in R is isomorphic to $\mathbb{Z}/s\mathbb{Z}$. We showed that we can compute the characteristic of R by computing the characteristic of $\langle 1_R \rangle$, so we get the following.

Corollary 27.5. *Let R be a commutative ring with unity. If R has characteristic $n > 0$, then it contains a subring isomorphic to $\mathbb{Z}_n$, and if it has characteristic 0, then it has a subring isomorpic to $\mathbb{Z}$.*

As a field cannot contain a subring with 0-divisors we get from the above corollary that it must have characteristic 0 or prime p, and so we get the following, (using that $\mathbb{Q}$ is the quotient field of $\mathbb{Z}$.)

Corollary 27.6. *A field F either has prime characteristic p and contains a subfield isomorphic to $\mathbb{Z}_p$, or has characteristic 0 and contains a subfield isomorpic to $\mathbb{Q}$.*

The fields $\mathbb{Z}_p$ and $\mathbb{Q}$ are are called *prime fields*.

Back to polynomial rings

Now we go back to the ring $F[x]$ of polynomials.

Theorem 27.7. *If F is a field, then every ideal in $F[x]$ is principle.*

Proof. Let N be an ideal in $F[x]$. If $N = 0$ or N contains a constant, then the result is trivial, so assume that N has a minimun degree element $g(x)$ with degree at least 1. If there is some $f(x)$ in N that is not in $\langle g(x) \rangle$ then the division algorithm finds us an element $r(x)$ of N of smaller degree than $g(x)$. $\square$

Theorem 27.8. *A principle ideal $\langle p(x) \rangle \neq \langle 0 \rangle$ of $F[x]$ is maximal if and only if $p(x)$ is irreducible.*

Proof. Clearly if $p(x)$ has a proper factor $a(x)$ then $\langle a(x) \rangle$ is an ideal properly containing $\langle p(x) \rangle$, so 'only if' is clear. Now assume that $p(x)$ is irreducible. If $\langle p(x) \rangle$ is not maximal then it is contained in some other proper ideal of $F[x]$; by the previous theorem it is principle, and as it is proper, it is not generated by a constant, so it is $\langle b(x) \rangle$ for some polynomial $b(x)$ of degree $n \geq 1$. So $b(x)$ divides $a(x)$, a contradiction. $\square$

We are going to use this a lot in the next section to extend fields F to fields E that contain zeros to the irredicble polynomials in $F[x]$. This would give us a way to prove unique factorisation. But actually, we can do it already; let's prove Theorem 23.13 which we used without proof in our proof of unique factorication in $F[x]$. It said that if irreducible $f(x)$ divides $a(x)b(x)$ then it divides one of them.

Proof. Assume that irreducible $f(x)$ in $F[x]$ divides $a(x)b(x)$. Then $a(x)b(x) \in \langle f(x) \rangle$ which is a maximal ideal as $f(x)$ is irreducible, and so prime by Corollary 27.3. But because $f(x)$ is prime, one of $a(x)$ or $b(x)$ is in it, meaning that $f(x)$ divides one of them. $\square$

Problems from the Text

Section 27: 3, 10-13, 14, 15-17, 20-23, 25, 31, 34-36

29 Extension Fields

We start this section with Kronecker's theorem.

Definition 29.1. A field E is an *extension* of a field F if it contains F as a subfield; we write $F \leq E$.

Theorem 29.2 (Kronecker's theorem)**.** *Let F be a field and $f(x)$ an irreducible polynomial in $F[x]$. There is an extension field E of F that contains a zero α of $f(x)$.*

A familiar example of Kronecker's theorem is that $f(x) = x^2 - 2$ is irreducible over $\mathbb{Q}$. There is an extension $\mathbb{R}$ that contains a zero $\sqrt{2}$ to $f(x)$. In fact there is a smaller extension inside $\mathbb{R}$: $\mathbb{Q}(\sqrt{2}) = \{p + \sqrt{2}q \mid p, q \in Q\}$.

What about with the polynomial $f(x) = x^3 + 3x + 2$ which we showed is irreducible in $\mathbb{Z}_5[x]$? As it is irreducible, $\langle f(x) \rangle$ is a maximal ideal in $\mathbb{Z}_5[x]$ so $E := \mathbb{Z}_5[x]/\langle f(x) \rangle$ is a field which properly contains $\mathbb{Z}_5$. Moreover the polynomial $f(x)$ that is irreducible over $\mathbb{Z}_5$ is no long irreducible over E, its zero is the element $\alpha := (x + \langle f(x) \rangle)$ of $\mathbb{Z}_5[x]/\langle f(x) \rangle$:

$$\begin{aligned} \alpha^3 + 3\alpha + 2 &= (x + \langle f(x) \rangle)^3 + 3(x + \langle f(x) \rangle) + (2 + \langle f(x) \rangle) \\ &= (x^3 + 3x + 2 + \langle f(x) \rangle) = (f(x) + \langle f(x) \rangle) = [0] \end{aligned}$$

The discussion above is basically the proof of Kronecker's theorem.

Problem 29.1. As α is a zero of x^3+3x+2 in $\mathbb{Z}_5[x]$, we get that $\alpha^3+3\alpha+2 = 0$, and so $\alpha^3 = 2\alpha+3$. Keeping this in mind, use polynomial long division to divide $x^3 + 3x + 2$ by $(x - \alpha)$. You get $(x - \alpha)g(x) = x^3 + 3x + 2$ for some degree 2 polynomial $g(x)$ with coefficients in $\mathbb{Z}_5(\alpha) = \{a + b\alpha + c\alpha^2 \mid a, b, c \in \mathbb{Z}_5\}$. Is $g(x)$ irreducible in $\mathbb{Z}_5(\alpha)[x]$? (You might want to use a computer to decide this.)

Our examples beg the following natural questions.

i). Are all extensions of a field F with a zero of $f(x)$ the same? Does this add the zeros of other irreducible polynomials?

ii). We know that all polynomials over $\mathbb{Q}$ factor into linear factors over $\mathbb{C}$. Is $\mathbb{C}$ the only/smallest extension of $\mathbb{Q}$ with this property? Can we build it in this way? How many times would we have to do this construction?

Definition 29.3. An element α of an extension E of F is *algebraic over* F if $f(\alpha) = 0$ for some $f(x) \in F[x]$; otherwise it is *transcendental over* F. An element of $\mathbb{C}$ is *algebraic* or *trancendental* if it is algebraic or transcendental over $\mathbb{Q}$.

Example 29.4. Though it is a bit tricky to show, π and e are transcendental over $\mathbb{Q}$. They are algebraic over $\mathbb{R}$ as π is a zero of the polynomial $f(x) = x - \pi$ in $\mathbb{R}[x]$.

Problem 29.2. Show that $\sqrt{1 + \sqrt{3}}$ is algebraic.

Observe that for an element α of an extension E of F, the evaluation homomorphism $\phi_\alpha : F[x] \to E$ is injective if and only if α is transcendental over E. Indeed if α is transcendental then the evaluation homomorphism $\phi_\alpha : F[x] \to E$ has kernel 0, while if it is algebraic then the kernal contains $\langle f(x) \rangle$ where $f(\alpha) = 0$ witnesses that α is algebraic.

Theorem 29.5. *If $\alpha \in E$ is algebraic over F then there is a unique monic irreducible polynomial $p(x)$ in $F[x]$ such that $p(\alpha) = 0$. It is the minimum degree polynomial $f(x)$ for which $f(\alpha) = 0$ and divides any such $f(x)$.*

Proof. The set N of polynomials of $F[x]$ that disappear at α is an ideal, as it is the kernel of the evaluation homomorphism ϕ_α to some extension field containing α, and so as F is a field it is principle: $N = \langle p(x) \rangle$ for some $p(x) \in F[x]$. We may assume that $p(x)$ is monic. If $f(x) = 0$ for some $f(x) \in F[x]$ then $f(x) \in N = \langle p(x) \rangle$, so $p(x) \mid f(x)$, and every other monic member of $\langle p(x) \rangle$ has degree greater than $p(x)$. If $p(x)$ is not irreducible, say if factors $p(x) = a(x)b(x)$ in to lower degree polymorphisms in $F[x]$, then as F is a field, $a(x) = 0$ or $b(x) = 0$ contradicting the fact that $p(x)$ is the minimum degree element of N. $\square$

Definition 29.6. For $\alpha \in E$ algebraic over F the monic irreducible polynomial $p(x)$ in $F[x]$ such that $p(\alpha) = 0$ is the *irreducible polynomial of α over F*. Its degree is *the degree of α over F*. We denote these as $\mathrm{irr}(\alpha, F)$ and $\deg(\alpha, F)$ respectively.

Problem 29.3. What are $\mathrm{irr}(\sqrt{2}, \mathbb{Q})$ and $\deg(\sqrt{2}, \mathbb{Q})$? What is $\mathrm{irr}(\sqrt{1+\sqrt{3}}, \mathbb{Q})$?

Definition 29.7. For an element α of an extension E of F, let the evaluation homomorphism $\phi_\alpha : F[x] \to E$ take F identically to F and x to α. The *simple extension $F(\alpha)$ of F by α* is the smallest subfield of E containing $\phi_\alpha[F[x]]$.

This field $F(\alpha)$ is a little different depending on whether α is algebraic or transcendental over F. If $\alpha \in E$ is algebraic over F, then $\phi_\alpha[F[x]]$ is isomorphic to $F[x]/\langle \mathrm{irr}(\alpha, F) \rangle$ so is a field, and $F(\alpha) = \phi_\alpha[F[x]]$. If α is transcendental over F, then $\phi_\alpha[F[x]]$ is an integral domain, and so $F(\alpha)$ is its quotient field.

Example 29.8. The ring we get from $\mathbb{Q} \leq \mathbb{R}$ by adjoining $\sqrt{2}$ is already a field; the element $\sqrt{2}$ already has an inverse $\sqrt{2}/2$. But the ring we get by adjoining π is not a field. The element π has no inverse (which we know from the external knowledge that π is trancendental).

Problem 29.4. Find the inverse of the root α of $f(x) = x^3 + 3x + 2$ in $\mathbb{Z}_5(\alpha)$.

Hint: It might be useful to write out the elements of $\mathbb{Z}_5(\alpha)$. As $\alpha^3 + 3\alpha + 2 = 0$ we have that $\alpha^3 = 2\alpha + 3$. So you can write every element P of $\mathbb{Z}_5(\alpha)$ as $P = a + b\alpha + c\alpha^2$ for $a, b, c \in \mathbb{Z}_5$. You might notice that you can represent P by a column vector $v_P = [a, b, c]^T \in \mathbb{Z}_5^3$. If v_P is $[a, b, c]^T$ what is $v_{(\alpha P)}$? You can express 'multiplication by α' as a matrix transformation: find the matrix M_α such that, for all $P \in \mathbb{Z}_5(\alpha)$, $M_\alpha v_P = v_{(\alpha P)}$. Then you can find the inverse x of α by solving $M_\alpha x = [1, 0, 0]^T$.

Theorem 29.9. *Let E be a simple extension of F, and $\alpha \in E$ be algebraic over F with degree $n \geq 2$. Then every element β of E can be written uniquely as $\beta = \sum_{i=0}^{n-1} b_i \alpha^i$ for elements $b_i \in F$.*

Proof. Any element β of E is $\beta = \phi_\alpha(f(x))$ for some $f(x) \in F[x]$. By the division algorithm there is unique $r(x)$ of degree less than $f(x)$ such that $f(x) = \text{irr}(\alpha, F)q(x) + r(x)$. So $\beta = \phi_\alpha(f(x)) = \phi_\alpha(r(x)) = r(\alpha)$ can be written as claimed.

For uniqueness, assume there is a second such representation. The difference of the two representations is a polynomial in α of degree at most $n-1$, and we know that this cannot be 0 as it is the image, under ϕ_α of a polynomial not in $\langle \text{irr}(\alpha, F) \rangle$. $\square$

Problems from the Text

Section 29: 5, 7, 17, 19-23, 29, 31, 33, 34

30 Vector Spaces

Recall that a real vector space was a set of vectors $V = \mathbb{R}^d = \{(v_1, \ldots, v_n) \mid v_i \in \mathbb{R}\}$ which we could add together component wise, and we could mutliply by scalars in $\mathbb{R}$. For a general vector space, we replace $V = \mathbb{R}^d$ with any abelian group A, and we replace the scalar field $\mathbb{R}$ with any field. (Seems like we are replacing everything! What's left?)

Definition 30.1. A *vector space V over a field F* is an abelian group with *scalar multiplication* $\cdot : F \times V \to V$ that satisfies the following properties for all $\alpha, \beta \in V$ and $a, b \in F$.

i). $a(b\alpha) = (ab)\alpha$

ii). $(a+b)\alpha = a\alpha + b\alpha$

iii). $a(\alpha + \beta) = a\alpha + a\beta$

iv). $1\alpha = \alpha$.

This is the main example of a vector space that we will be using:

Example 30.2. An extension E of a field F is a vector space over F. The additive group of E is the vector fields additive group. The multiplication $f\alpha$ of $\alpha \in E$ by $f \in F$ is the regular multiplication in E.

Most of the things that we need to define and prove about vector spaces we have done for real vector spaces in Linear Algebra. The proofs are basically the same. We need only one result.

In Theorem 29.9 we saw that if α is algebraic over F with degree $n \geq 2$ then any element of $F(\alpha)$ can be written uniquely as a linear combination (over F) of the elements in $A = \{1, \alpha, \ldots, \alpha^{n-1}\}$.

Viewing $F(\alpha)$ as a vector space over F, this means that the set A is a *basis* of $F(\alpha)$. Proving the general fact about vector space that the sizes of all bases are the same, we get that the elements of $B = \{1, \beta, \ldots, \beta^n\}$ cannot be *independent*. So there is a linear combination of the elements of B equalling 0, which defines a polynomial of degree at most n that disappears on β.

Theorem 30.3. *Let α be algebraic of degree n over F. Any element β of $F(\alpha)$ is algebraic of degree at most n over F.*

Problems from the Text

Section 30: -

31 Algebraic Extensions

An extension E of a field F is an *algebraic extension of* F if every element of E is algebraic over F. If it has finite dimension n as a vector space over F then it is a *finite extension of degree* n. We denote this degree by $[E\colon F]$. If E is a finite extension of F of degree n, then for any $\beta \in E$ the elements $\{1, \beta, \ldots, \beta^n\}$ are dependent, and so β is algebraic over F. Thus we get the following.

Fact 31.1. *Any finite extension E of F is an algebraic extension.*

The following comes from a simple argument that for a basis B of E over F and a basis A of F over K, AB is a basis of E over K.

Theorem 31.2. *Let K be a finite extension of E and let E be a finite extension of F The field K is a finite extension of F and*

$$[K : F] = [K : E][E : F].$$

The following is a useful corollary of this theorem.

Problem 31.1. Show that if α is algebraic over F and $\beta \in F(\alpha)$ then $\deg(\beta, F) \mid \deg(\alpha, F)$.

Example 31.3. As $\sqrt{2}$ has degree 2 over $\mathbb{Q}$, and any zero of $x^3 - 2$ has degree (dividing) 3 over $\mathbb{Q}$, $\sqrt{2}$ cannot be a zero of $x^3 - 2$.

Let $\alpha_1, \ldots, \alpha_n$ be a set of elements of an extension field E of F. The field $F(\alpha_1, \alpha_2, \ldots, \alpha_n)$ is the smallest extension field of F that contains all of the α_i. We call it the field F *with the elements α_i adjoined*. It can also be defined as the intersection of all extensions of F containing the α_i. And so it is clear that one can build it as $[\ldots[F(\alpha_1)](\alpha_2)]\ldots(\alpha_n)]$, and that the order in which the α_i are adjoined does not matter.

Example 31.4. We know that $(1, \sqrt{2})$ is a basis of $\mathbb{Q}(\sqrt{2})$ over $\mathbb{Q}$. And we can verify that $\mathrm{irr}(\sqrt{2}+\sqrt{3}, \mathbb{Q}) = x^4 - 10x^2 + 1$. So $[\mathbb{Q}(\sqrt{2}+\sqrt{3}) : \mathbb{Q}] = 4$. It follows that $\sqrt{2}+\sqrt{3} \notin \mathbb{Q}(\sqrt{2})$ and so neither is $\sqrt{3}$. Thus $(1, \sqrt{3})$ is a basis of $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ over $\mathbb{Q}(\sqrt{2})$. We get by the proof of Theorem 31.2 (which you found in the text) that $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ has basis $(1, \sqrt{2}, \sqrt{3}, \sqrt{6})$ over $\mathbb{Q}$.

Example 31.5. One can show that $2^{1/2}$ is not in $\mathbb{Q}(2^{1/3})$ because it has degree $\deg(2^{1/2}, \mathbb{Q}) = 2$ which does not divide $\deg(2^{1/3}, \mathbb{Q}) = 3$. So $[\mathbb{Q}(2^{1/2}, 2^{1/3}) : \mathbb{Q}] = 6$. It is also easy to see that $\mathbb{Q}(2^{1/2}, 2^{1/3}) \subset \mathbb{Q}(2^{1/6})$ which has degree at most 6 over $\mathbb{Q}$. So the fields are equal, and so $x^6 - 2$ is irreducible over $\mathbb{Q}$.

The above example shows that even though it does not look like it, an extension $F(\alpha_1, \alpha_2, \ldots, \alpha_n)$ might be simple.

We have seen that any finite extension E of F is algebraic. But we can say more.

Theorem 31.6. *An extension E of F is finite if and only if it can be expressed as an extension $F(\alpha_1, \alpha_2, \ldots, \alpha_n)$ of F by finitely many algebraic elements α_i.*

Proof. The 'if' part of this theorem is obvious. The only if is only a little bit less obvious. The point is that if some extension $F' := F(\alpha_1, \alpha_2, \ldots, \alpha_m)$ of F in E is not all of E, then there is some algebraic element of E that we can attach to it, and this, having degree at least 2, reduces the differences in the degrees of $[[E|F]$ and $[F'|F]$. As $[E|F]$ is finite, this need be done only finitely many times. □

Algebraic Closure

In this part, we look at some related ideas that have almost the same name, so are maybe easy to confuse.

Definition 31.7. Let E be an extension of a field F.

- The *algebraic closure of F in E* is the set
$$\overline{F}_E = \{\alpha \in E \mid \alpha \text{ is algebraic over } F\}.$$
- The field F is *algebraically closed* if every non-constant polynomial in $F[x]$ has a zero in F.
- E is an algebraic closure of F if it is algebraically closed.

It is not hard to see that $\overline{F}_E$ is, itself, a field, indeed, for α and $\beta \in E$ algebraic over F, $F(\alpha, \beta)$ contained in E is a finite extension of F, so all its elements, including $\alpha + \beta$, $\alpha\beta$, and α^{-1} are algebraic over F. Thus they are in $\overline{F}_E$, and so it is closed under both operations and has inverse. As the set of all algebraic numbers is the algebraic closure of $\mathbb{Q}$ in $\mathbb{C}$, they are a field, and indeed are then an algebraic closure of $\mathbb{Q}$.

Problem 31.2. Show that a field F is algebraically closed if and only if every non-constant polynomial in $F[x]$ factors into polynomials of degree 1. Conclude that no algebraically closed field has any non-trivial algebraic extensions.

We get that $\mathbb{C}$ is algebraically closed from the following theorem which you have likely seen before. The proof in the book is analytic, and, ironically, considering the title of the theorem, I could only find analytic proofs of it. We skip the proof.

Theorem 31.8 (Fundamental Theorem of Algebra)**.** *Every polynomial in $\mathbb{C}[x]$ has a zero in $\mathbb{C}$, so factors into linear factors in $\mathbb{C}[x]$.*

The following seems easy: just keep attaching algebraic elements until you have zeros to all polynomials, but infinity is a tricky beast, and we have to use Zorn's Lemma to prove it.

Theorem 31.9. *Every field has an algebraic closure.*

Proof. Let Ω be the set of all zeros of all polynomials over a field F. All algebraic extensions of F are subsets of Ω, so are partially ordered by inclusion. For a chain

$$T : F_{i_1} \subset F_{i_2} \subset \dots$$

in this partial order we claim that $W_T = \cup_{i_j} F_{i_j}$ is an algebraic extension of F containing all extensions in the chain. To see that it is algebraic, it is enough to note that any element is in one of the F_{i_j} so has finite degree. To see that W_T is a field, we observe for α and β that there is some j large enough that F_{i_j} contains both, and so contains $\alpha + \beta$ and $\alpha\beta$. So W_T does as well.

Every chain is bounded, and so Zorn's Lemma says there is some maximal element $\overline{F}$ in the ordering. If this is not algebraically closed, then we can attach another algebraic element, contradicting the fact that it is maximal in the ordering. $\square$

Note that defining Ω as I have is cheating– I should use the axiom of choice for this too. Look in the book for a more rigourous treatment of this. I find it confusing.

Our proof suggests that there may be several algebraic closures of F, while our notation $\overline{F}$ suggests that there is only one. There can be many, but one can show that they are isomorphic, so we will talk of *the* algebraic closure $\overline{F}$ of F.

Problems from the Text

Section 31: 1-13, 14-18, 19, 22, 25, 26, 30, 33, 35

33 Finite Fields

We have seen that any finite extension of a field is algebraic, so any finite field F is an algebraic extension of its prime subfield $\mathbb{Z}_p$. As it is a vector space over $\mathbb{Z}_p$, it has p^n elements for some n. As the multiplicative group F^* has order $p^n - 1$, we have $\alpha^{p^n} = \alpha$, so the elements of F are exactly the zeros, in $\overline{\mathbb{Z}}_p$ of $x^{p^n} - x \in \mathbb{Z}_p[x]$.

On the other hand, Galois showed the following.

Theorem 33.1. *For any prime p and integer $n \geq 1$, there is a finite field of order p^n.*

Proof. As $\overline{\mathbb{Z}}_p$ is algebraically closed, $P(x) = x^{p^n} - x$ has p^n zeros. We show that the set F of zeros are distinct, and are the elements of a field. Clearly 0 is a zero of $P(x)$ of multiplicity 1. We show that any zero α of $P^*(x) = x^{p^n-1} - 1$ has multiplicity 1. Indeed as α is a zero, we have

$$P^*(x) = (x - \alpha)(x^{p^n-2} + \alpha x^{p^n-3} + \ldots \alpha^{p^n-3}x + \alpha^{p^n-2}) =: (x - \alpha)g(x),$$

and plugging α into $g(x)$ we get

$$g(\alpha) = (p^n - 1)\alpha^{p^n-2} = -\alpha^{p^n-2} = -\frac{\alpha^{p^n-1}}{\alpha} = -\frac{1}{\alpha} \neq 0.$$

Now 0 and 1 are in F, and $|F| = p^n$. It is enough to show that F is closed under our operations, by showing that $P(\alpha + beta) = 0$ and $P(\alpha\beta) = 0$. From this the calculation above shows that $\alpha \neq 0$ has mutliplicative inverse $-g(\alpha) \in F$.

Closed under mutliplication: $(\alpha\beta)^{p^n} = \alpha^{p^n}\beta^{p^n} \equiv_p \alpha\beta$, so $P(\alpha\beta) = 0$. .

Closed under addition: Observe that $(\alpha + \beta)^{p^n} \equiv_p \alpha + \beta$. This is true by the binomial theorem for $n = 1$ and then by induction by observing that

$$(\alpha + \beta)^{p^i} = [(\alpha + \beta)^{p^{i-1}}]^p \equiv_p (\alpha + \beta)^p \equiv_p \alpha + \beta.$$

□

Now we saw in Chapter 23 that the multiplicative group of a field is cyclic. So for a finite field F of order p^n, F^* is cyclic, and so has a generator α. It follows that F is generated by α over any subfield K so is a simple extension $K(\alpha)$ of K.

Corollary 33.2. *Any finite extension E of a finite field F is a simple extension.*

In particular any finite filed F is a simple extension $\mathbb{Z}_p(\alpha)$ of its prime subfield. For any two finite fields F and F' of order p^n there is a isomorphism

$\phi : F \to F'$ taking the prime field $\mathbb{Z}_p$ to $\mathbb{Z}_p$ and the generator α of the first to the generator α' of the other. So upto isomorphism, there is a unique field of order p^n. It is called the *Galois Field* $\mathrm{GF}(p^n)$.

As the non-zero elements $\mathrm{GF}^*(p^n)$ of the Galois Field of order p^n are roots of $x^{p^n-1} = 1$. They are called the $(p^n-1)^{th}$ *roots of unity*. Any generator α of $\mathrm{GF}^*(p^n)$ is a *primitive* $(p^n-1)^{th}$ *root of unity*, and generates $\mathrm{GF}(p^n)$ as a simple extension $\mathbb{Z}_p(\alpha)$ of the prime field $\mathbb{Z}_p$.

Corollary 33.3. *If F is a finite field, then for every $n \geq 1$, there is a irreducible polynomial in $F[x]$ of degree n.*

Proof. Where F has characteristic p it is $\mathrm{GF}(p^d)$ for some d, so is the extension of $\mathrm{GF}(p)$ consisting of the zeros of $x^{p^d} - x$. Its elements α satisfy $\alpha^{p^d} = \alpha$ so satisfy

$$\alpha^{(p^d)^n} = \alpha^{(p^d)^{n-1}} = \cdots = \alpha^{(p^d)} = \alpha$$

So all are elements of $\mathrm{GF}(p^{dn})$, which consists of the zeros of $x^{p^{dn}} - x$. By the previous corollary $\mathrm{GF}(p^{dn})$ is a simple extension of F so the irreducible polynomial of its generator has degree $[\mathrm{GF}(p^{dn}) : \mathrm{GF}(p^d)] = n$. □

Problems from the Text

Section 33: 2,5,8,9,10,13,14

34 Isomorphism Theorems

The Homomorphism Theorem for groups that we saw in Chapter 14, is often called the First Isomorphism Theorem, it is grouped together with the two other Isomorphisms Theorems. Before we introduce the other two, lets recall the first. We write it again multiplicatively– we will be applying these results to groups of automorphisms of fields, and these are usually written multiplicatively.

Theorem 34.1 (First Isomorphism Theorem)**.** *A group homorphism $\phi : G \to G'$ with kernel N decomposes at $\phi = \mu \circ \gamma_K$ where*

$$\gamma : G \to G/N : g \mapsto gN \qquad \text{and} \qquad \mu : G/N \to \phi(G) : gN \mapsto \phi(g)$$

are the quotient homomorphism and an isomorphism respectively.

Not only does the quotient γ map G to G/N, it maps the normal subgroups of G to those of G/N.

Lemma 34.2. *Let N be a normal subgroup of G, and γ be the canonical quotient $\gamma : G \to G/N$. The map $L \mapsto \gamma[L]$ is a well defined bijection from the set of normal subgroups of G that contain N to the set of normal subgroups of G/N.*

Proof. Theorem 15.16 of the text tells us that if L is a normal subgroup of G containing N then $\gamma[L] \subset G/N$ is a normal subgroup, so the map is well defined on the domain and codomain claimed. To see that is injective, assume that $\gamma[L'] = \gamma[L]$. In particular, for $\ell' \in L'$, we have that $\gamma(\ell') \in \gamma[L'] = \gamma[L]$ so for some $\ell \in L$ we have $\ell' N = \gamma(\ell') = \gamma(\ell) = \ell N$. So $\ell' = \ell n$ for some $n \in N$, but ℓ and n are both in L, so $\ell \in L$.

To see that it is surjective, let H be a normal subgroup of G/N, and let $L = \gamma^{-1}[H]$, so clearly $\gamma[L] = H$. This set L is a normal subgroup of G by Theorem 15.16, and contains N as $\gamma[N] = N$ is the identity of G/N.

□

Let H and N be two subgroups of a group G. The set $HN = \{hn \mid h \in H, n \in N\}$ need not be a group, as it is not necessarily closed, but...

Lemma 34.3. *For subgroups H and N of G, if N is normal then $HN = NH$ is a subgroup of G. If H is also normal then HN is normal in G.*

When it is a group, HN is called the *join* of H and N. It contains both of N and H as subgroups.

Problem 34.1. Prove the lemma.

Theorem 34.4 (Second Isomorphism Theorem)**.** *Let H and N be subgroups of G. If N is normal, then $(HN)/N \cong H/(H \cap N)$.*

Proof. Let γ be the canonical quotient $\gamma : G \to G/N$. Observe that the images $\gamma[HN]$ and $\gamma[H]$ are the same subgroup of G/N as $\gamma(hn) = hnN = hN = \gamma(h)$.

Consider the restriction $\gamma_H : H \to G/N$ of γ to H, it is onto $\gamma[H]$. Its kernel consists of the element of H in $N = \ker(\gamma)$, so is $H \cap N$. By the first isomorphism theorem we get then that $\gamma[H] \cong H/(H \cap N)$.

Now consider the restriction $\gamma_{HN} : HN \to G/N$, which is onto $\gamma[HN] = \gamma[H]$. Its kernel is N, and so $\gamma[HN] \cong HN/N$. □

If $K \leq H$ are normal subgroups of G then H/K is a normal subgroup of G/K. Indeed as H is normal, we have for $g \in G$ and $h \in H$ that $gh = h'g$ for some h' in H. So $(gK)(hK) = (h'K)(gK)$, and so $(gK)(H/K) = (H/K)(gK)$. As this was for any gK, H/K is normal in G/K.

Theorem 34.5 (Third Isomorphism Theorem)**.** *Let H and K be normal subgroups of G. If $K \leq H$ then*

$$G/H \cong (G/K)/(H/K).$$

Proof. Show that $\phi : G \to (G/K)/(H/K) g \mapsto (gK)(H/K)$ is an onto homomorphism with kernel H; the results then follows by the first homomorphism theorem. □

Problems from the Text

Section 34: 1,3,5,7,8

35 Series of Groups

A sequence $H_0, H_1, \ldots, H_n$ of subgroups of G is a *subnormal series* of G if (it is finite) and H_i is a normal subgroup of H_{i+1} for each i. It is a *normal series* of G if, further, each of the H_i are normal in G.

Example 35.1. Recall the (non-abelian) dihedral group D_4 of symmetries of a square. It was generated by a rotation ρ by 90 degrees, and by a flip μ around a vertical bisector. The series

$$\{1\} < \{1, \mu\} < \{1, \rho^2, \mu, \mu\rho^2\} < D_4$$

is a subnormal series: the subgroup $H_2 = \{1, \rho^2, \mu, \mu\rho^2\}$ is normal as it has index 2 in D_4 so there are only two cosets, we can check for any element α not in H_2 that αH_2 an $H_2\alpha$ are the other coset; as H_2 is abelian, all others 'steps' in the series are normal. The series is not normal though as $H_1 = \{1, \mu\}$ is not normal in D_4: $\mu\rho = \rho^3\mu$ so $\rho H_1 = \{\rho, \rho\mu\}$ while $H_1\rho = \{\rho, \rho^3\mu\}$.

Normal and subnormal series of G are the same if G is abelian, and not really interesting in this case, but we may use abelian examples to illustrate ideas.

A subnormal series $\{K_i\}$ is a *refinement* of a subnormal series $\{H_i\}$ of G if $\{H_i\} \subset \{K_i\}$.

Example 35.2. $\{1\} < \{1, \mu\} < \{1, \rho^2, \mu, \mu\rho^2\} < D_4$ is a refinement of $\{1\} < \{1, \rho^2, \mu, \mu\rho^2\} < D_4$.

The point of finding a subnormal series of a group is that it defines a set $\{H_{i+1}/H_i\}$ of factor groups that tell us a lot about G.

Two subnormal series $\{H_i\}$ and $\{K_i\}$ of a group are *isomorphic* if the sets $\{H_{i+1}/H_i\}$ and $\{K_{i+1}/K_i\}$ are the same, upto isomorphism of the groups. (These are sets, so order doesn't matter.)

Example 35.3. The two normal series'

$$\{0\} < \langle 5\rangle < \mathbb{Z}_{15} \qquad \text{and} \qquad \{0\} < \langle 3\rangle < \mathbb{Z}_{15}$$

are isomorphic as the factor groups of the first are $\{\mathbb{Z}_{15}/\langle 5\rangle \cong \mathbb{Z}_3, \langle 5\rangle/\{0\} \cong \mathbb{Z}_5\}$ and of the second are $\{\mathbb{Z}_{15}/\langle 3\rangle \cong \mathbb{Z}_5, \langle 3\rangle/\{0\} \cong \mathbb{Z}_3\}$.

Our goal is to find a canonical subnormal series for a group. Clearly we can take this by using the normal series $\{1\} < G$, but this is not a useful series. A

good first step is to find a 'longest', or most refined, series, to show that if this exists, (and according to our definition of longest, it won't always) all of these longest series have the same factor groups.

The Schreier Theorem

A useful tool in showing that there is a canonical refinement of all series of G is a result of Schreier that says that any two subnormal series have isomorphic refinements.

Problem 35.1. Show that the normal series

$$\{0\} < 8\mathbb{Z} < 4\mathbb{Z} < \mathbb{Z} \qquad \text{and} \qquad \{0\} < 9\mathbb{Z} < \mathbb{Z}$$

have isomorphic refinements.

Theorem 35.4. *Any two subnormal series of a group G have isomorphic refinements.*

Proof. Let G have subnormal series

$$\{1\} = H_0 < H_1 < H_2 < \cdots < H_n = G \qquad \{1\} = K_0 < K_1 < K_2 < \cdots < K_m = G.$$

We will define a refinement

$$\begin{array}{ccccccccc}
H_0 & = & H_{0,0} & < & H_{0,1} & < \cdots < & H_{0,m} & = & H_1 \\
H_1 & = & H_{1,0} & < & H_{1,1} & < \cdots < & H_{1,m} & = & H_2 \\
 & \vdots & & & & & \vdots & & \\
H_{n-1} & = & H_{n-1,0} & < & H_{n-1,1} & < \cdots < & H_{n-1,m} & = & H_n
\end{array}$$

of the first and

$$\begin{array}{ccccccccc}
K_0 & = & K_{0,0} & < & K_{0,1} & < \cdots < & K_{0,n} & = & K_1 \\
K_1 & = & K_{1,0} & < & K_{1,1} & < \cdots < & K_{1,n} & = & K_2 \\
 & \vdots & & & & & \vdots & & \\
K_{m-1} & = & K_{m-1,0} & < & K_{m-1,1} & < \cdots < & K_{m-1,n} & = & K_m
\end{array}$$

of the second such that $H_{i,j+1}/H_{i,j} \cong K_{j,i+1}/K_{j,i}$ for all i and j. As every '$<$' corresponds to a factor group of the two groups it compares, that there are the same number in the two refinements means that we have the same number of factor groups. The condition $H_{i,j+1}/H_{i,j} \cong K_{j,i+i}/K_{j,i}$ means that those in the ith row of them in the first refinement are isomorphic to those in the ith column of them in the second refinement.

We do this as follows: Let $H_{i,j} = H_i(H_{i+1} \cap K_j)$ and let $K_{i,j} = K_i(K_{i+1} \cap H_j)$. As $K_0 = \{1\}$ and $H_n = G$ it is easy to check that $H_{i,0} = H_i$ and $H_{i,m} = H_{i+1}$. So it is enough to prove the following Lemma. $\square$

Lemma 35.5 (Zassenhaus Lemma)**.** *Let H and K be subgroups of G with normal subgroups H^* and K^* respectively. The group $H^*(H \cap K^*)$ is a normal subgroup of $H^*(H \cap K)$, and $K^*(K \cap H^*)$ of $K^*(K \cap H)$, and*

$$H^*(H \cap K)/H^*(H \cap K^*) \cong K^*(K \cap H)/K^*(K \cap H^*).$$

Proof. To prove this we show

i). $L := (H^* \cap K)(H \cap K^*)$ is a normal subgroup of $(H \cap K)$,

ii). $H^*(H \cap K^*)$ is a normal subgroup of $H^*(H \cap K)$, and

iii). $H^*(H \cap K)/H^*(H \cap K^*) \cong (H \cap K)/L$.

The result then follows by the fact that H and K are interchangable.

For part i). it is enough, by Lemma 34.3 to show that $H^* \cap K$ is a normal subgroup of $H \cap K$; This is exercise 22 of this chapter in the text.

Part ii) and iii) come from the first isomorphism theorem by showing that the map

$$\phi : H^*(H \cap K) \to (H \cap K)/L$$

defined by $\phi(hx) = xL$ for $h \in H^*$ and $x \in H \cap K$ is a surjective homomorphism with kernel $H^*(H \cap K^*)$. The main part is to show that ϕ is a well defined homomorphism. Then that it is surjective is clear, and that the kernel is $H^*(H \cap K^*)$ is clear when we recall that $L = (H^* \cap K)(H \cap K^*)$.

So see that ϕ is well defined, assume that if $h'x' = hx$ for some other $h' \in H^*$ and $x' \in H \cap K$. We must show that $x'L = \phi(h'x') = \phi(h, x) = xL$. Well, this is true, because $x'x^{-1}$ is in $H \cap K$ and $x'x^{-1} = h'^{-1}h$ is in H^*, so it is in $H^* \cap K \subset L$.

That ϕ is a homomorphism uses that H^* is normal in H. This gives us that $xh' = h^*x$ for some $h^* \in H^*$ and so

$$\phi((hx)(h'x')) = \phi(hh^*xx') = xx'L = (xL)(x'L) = \phi(hx)\phi(h'x)$$

as needed. □

The Jordan-Hölder Theorem

To get a canonical longest subnormal series of a group, we should refine it as much as we can.

A subnormal series $\{H_i\}$ of a group G is a *composition series* if all of the factor groups H_{i+1}/H_i are simple. A composition series that is normal is a *principle series*.

Recall that H_{i+1}/H_i is simple if and only if H_i is a maximal normal subgroup of H_{i+1}, and this always exists if H_{i+1} is not simple. Thus for any finite $G = H_n$, we can make a finite composition series by finding a maximal normal subgroup H_{n-1} of H_n, and then H_{n-2} of H_{n-1} etceter.

Fact 35.6. *Any finite group G has a finite composition series.*

Example 35.7. For any n the alternating group A_n is normal in the symmetric group S_n, so

$$\{0\} < \langle A_n \rangle < S_n$$

is normal. If $n \neq 4$ then A_n is simple, so the series is principle.

Finiteness is necessary for this though.

Example 35.8. The group $\mathbb{Z}$ has no composition series. Indeed, in any (sub)normal series

$$\{0\} < H_1 < \cdots < H_n$$

the smallest non-trivial group H_1 is $m\mathbb{Z}$ for some m, and this is isomorphic to $\mathbb{Z}$, so has many normal subgroups.

Example 35.9. Both of the normal series

$$\{0\} < \langle 5 \rangle < \mathbb{Z}_{15} \qquad \text{and} \qquad \{0\} < \langle 3 \rangle < \mathbb{Z}_{15}$$

of $\mathbb{Z}_{15}$ are composition series, indeed, they are principle series.

Moreover, these series are isomorphic. They must be, by Theorem 35.4 they have isomorphic refinements, but they are composition series so they have no proper refinements. This argument gives the following.

Theorem 35.10 (Jordan-Hölder Theorem)**.** *Any two composition series of a group G are isomorphic.*

While to find a composition series of a group H_n, it is enough to find any maximal normal subgroup H_{n-1}. To find a maximal normal H_{n-2} of H_{n-1}, etc, this might be difficult. But perhaps there are some obvious normal subgroups. They can help.

Theorem 35.11. *If G has a composition series, and N is a proper normal subgroup of G, then there is a composition series containing N.*

Proof. The series $\{e\} < N < G$ is normal, so has a refinement isomorphic to any composition series of G. □

Definition 35.12. A group G is *solvable* if it has a composition series $\{H_i\}$ in which all factor groups H_i/H_{i-1} are abelian.

Example 35.13. The group S_3 has composition series $\{e\} < A_3 < S_3$ with factor groups isomorphic to the abelian groups $\mathbb{Z}_3$ and $\mathbb{Z}_2$, so S_3 is solvable. The group S_5 has composition series $\{e\} < A_5 < S_5$ with factor groups A_5 and $\mathbb{Z}_2$. A_5 is simple but not abelian. So S_5 is not solvable. Indeed neither is A_5; it is the smallest non-solvable group.

Problems from the Text

Section 35: 2, 5, 6, 8, 12, 15-17, 18, 22, 26, 28

36 Sylow Theorems

For any number d dividing the order $|G|$ of an abelian group, there is a subgroup of G of order d. This does not hold for non-abelien groups, as the alternating group A_4 has order 12, but no subgroup of order 6. We see though, among other things, that if $d = p^n$ is prime power dividing the order of G, then it has a subgroup of order d.

A set X is a *G-set* if it has a *G-action*– a map $\phi : G \times X \to X : (g, x) \mapsto g * x$ such that $ex = x$ for all $x \in X$ and $(gg')x = g(g'x)$ for all $g, g' \in G$ and $x \in x$.

Example 36.1. G itself is a G-set with G-action simply defined $g*x = gx$. The group $G \times G \times G$ is a G-set with G action defined by $g * (a, b, c) = (ga, gb, gc)$.

The set $G * x \subset X$ is the *orbit* of x under the G-action $*$.

Problem 36.1. Show that if $y \in G * x$ then $G * y = G * x$. Conclude that the orbits of a G-action on X partition X.

Problem 36.2. For G-set X, fix x in X. Let $G_y = \{g \in G \mid gx = y\}$, for any $y \in G * x$. Show that

i). the set G_x is a subgroup of G, and

ii). the other sets G_y for $y \in G * c$ are the left cosets of G_x.

Conclude that $|G * x| = (G : G_x)$.

Let X_G be the set of elements of X that are fixed by all of G– the set of elements in one element orbits.

Theorem 36.2. *For any group G of order p^n and any G-set X, $|X_G| \equiv_p |X|$.*

Proof. Let $\{x_1, \ldots, x_r\}$ be a set of representitives of the orbits of X, and assume wlog that $X_G = \{x_1, \ldots, x_s\}$, where $s \leq r$. As the orbits of X partition X we

have that

$$|X| = \sum_{i=1}^{r} |G * x_i| = |X_G| + \sum_{i=s+1}^{r} |G * x_i|.$$

But each $|G * x_i|$ divides p^b, and as it is bigger than 1 it is divisible by p. So $|X_G| \equiv_p |X|$. $\square$

A group G is a *p-group*, for a prime p, if every element has order that is a power of p. A subgroup of a group G is a *p-subgroup* of G if it is a p-group.

Theorem 36.3 (Cauchy's Theorem)**.** *For prime p dividing $|G|$, G has an element of order p, so has a subgroup of order p.*

Proof. Let X be the set of ordered p-tuples $(g_1, g_2, \ldots, g_p)$ of elements of G such that $g_1 g_2 \ldots g_p = e$, and define a $\mathbb{Z}_p$-action on X by letting $1 * (g_1, g_2, \ldots, g_p) = (g_2, g_3, \ldots, g_p, g_1)$. The fixed points $X_{\mathbb{Z}_p}$ of this action are the constant tuples, so yields elements g of G such that $g^p = e$. We know that $|X| = |G|^{p-1}$, as after choosing the first $p-1$ elements in a tuple, the last is uniqely determined. So $|X| \equiv_p 0$. But $|X_{\mathbb{Z}_p}| \equiv_p |X|$ and $X_{\mathbb{Z}_p}$ contains the element $(e, e \ldots, e)$ so it must contain some other tuple $(g, g \ldots, g)$, which means $g^p = e$ as needed. $\square$

Corollary 36.4. *G is a p-group if and only if $|G|$ is a power of p.*

The normaliser of a subgroup

For a group G let $\mathscr{S}$ be the set of subgroups of G. For g in G and $S \in \mathscr{S}$ the set gSg^{-1} is easily seen to be a subgroup of G. So the action $g * S = gSg^{-1}$ is a G-action on $\mathscr{S}$. It is called *conjugation* by g. The subgroup gSg^{-1} of G is *conjugate* to S.

Problem 36.3. Consider the conjugate G-action on $\mathscr{S}$. Show that H is a normal subgroup of the subgroup G_H of G that fixes H.

We call G_H the *normaliser* of H, and denote it $N[H]$.

The following are Lemma's 36.6 and Corollary 36.7 of the text.

Problem 36.4. Let H be a p-subgroup of a finite group G.

i). Show that $(N[H] : H) \equiv_p (G : H)$.

ii). Conclude that if p divides $(G : H)$ then $N[H] \neq H$.

The Sylow Theorems

Theorem 36.5 (First Sylow Theorem)**.** *For a finite group G of order $p^d m$ where p does not divide m, G contains a subgroup of order p^i for each $i \leq d$, and if $I < d$, then this is normal in a subgroup of order p^{i+1}.*

Proof. We have by Cauchy that G contains a subgroup of order p; proceeding by induction assume that it contains a subgroup H of order p^i for $i < d$ we show that it contains a subgroup of order p^{i+1}. Indeed, we have that $p|mp^{d-i} = (G : H)$ so $N[H] \neq H$ and $p|(N[H] : H)$. As H is normal in $N[H]$, the factor $N[H]/H$ is a group, and p divides its order. By Cauchy it has a subgroup of order p which pulls back, via the factor homomorphism, to a subgroup H' of $N[H]$ containing H which has order p^{i+1}. As H was normal in $N[H]$ it is normal in the subgroup H'. □

Problem 36.5. Use the First Sylow Theorem to show that any group G with prime power order $|G| = p^d$ is solvable.

A *Sylow p-subgroup* of a group G is a maximal p-subgroup of G; when G has order $p^d m$ then it is a subgroup of order p^d.

Theorem 36.6 (Second Sylow Theorem)**.** *Any two Sylow p-subgroups of a finite group G are conjugate.*

Proof. Assume that G has two Sylow p-subgroups P_1 and P_2, and let P_2 act on the set of cosets of P_1 by $y*(xP_1) = yxP_1$. The number of singleton orbits is the same mod p as the number m of cosets of P_1, so some coset xP_1 is fixed by the action, and so $yxP_1 = xP_1$ for all $y \in P_2$. But then $x^{-1}yxP_1 = x^{-1}xP_1 = P_1$ for all $y \in P_2$ and so $x^{-1}P_2x \subset P_1$. As $|P_1| = |P_2|$ this containment is equality, and P_1 and P_2 are conjugate. □

Theorem 36.7 (Third Sylow Theorem)**.** *If p divides $|G|$ then the number of Sylow p-subgroups of G divides $|G|$ and is congruent to 1 modulo p.*

Proof. See text. □

While the classification of abelian groups was pretty easy, doing the same for non-abelian groups is hard. Indeed, even classifying all of the finite simple groups was an immense project consisting of 100s of papers and was only finished in the early 2000s.

The Sylow theorems were a great tool in the classification, and we see now how they can be used.

Example 36.8. We show that there is no simple group of order 15, by showing that any group G of order 15 has a normal subgroup of order 5. The first Sylow theorem tells us there is a subgroup of order 5, and the third tells us there is

only one, call it H. As gHg^{-1} is a subgroup of G for any g, we have that $gHg^{-1} = H$ for all g, and so H is normal.

In fact the proof of this example shows us that any group G of order pq for primes $p < q$ has a normal subgroup of order p. If $q \equiv_p 1$ then it also has a normal subgroup of order q, and it is not too hard to show that $G = \mathbb{Z}_p \times \mathbb{Z}_q$.

This example is from Chapter 37.

Example 36.9. There is no simple subgroup of order $30 = 2 \cdot 3 \cdot 5$. Indeed Sylow tells us the possible number of 3-subgroups is 1 or 10 and the possible number of Sylow-5 subgroups is 1 or 6. If there is only one of either of these, then it is normal by the above arguments, so we may assume that there are 6 Sylow 5-subgroups. These are all conjugate, and their intersections are subgroups of them, so they can only intersect in the identity element. We thus have $4 \times 6 = 24$ elements that have order 5. With the identity of order 1 this leaves only 5 elements that can have order other than 1 or 5. A similar argument for 3-subgroups says we must have at least 20 elements of order 3 which is too many elements.

Problems from the Text

Section 36: 1, 3, 5, 7-10, 11, 12, 13, 14, 20

48 Automorphisms of fields

In the next chapter we will show that all algebraic closures of a field F are isomorphic, and so from now, we will assume that all elements that are algebraic over F are in some arbitrary algebraic closure $\overline{F}$.

Let E be an algebraic extension of a field F. Two elements α and β of E are *conjugate over* F if $\mathrm{irr}(\alpha, F) = \mathrm{irr}(\beta, F)$. Observe that if β is a zero of $f(x) = \mathrm{irr}(\alpha, F)$ then $\mathrm{irr}(\beta, F)$ is $f(x)$; it certainly must divide it, and if it is not equal to it, then $f(x)$ is not irreducible. So two elements of E are conjugate over F if they are zeros of the same irreductible polynomial over F.

Example 48.1. The roots i and $-i$ of the polynomial $\mathrm{irr}(i, \mathbb{R}) = \mathrm{irr}(-i, \mathbb{R}) = x^2 - 1$ in $\mathbb{R}[x]$ are conjugate.

Good, right? They deserve it– in your complex analysis class you called them complex conjugates! Maybe you have seen that $\phi : a + bi \mapsto a - bi$ is an isomorphism of $\mathbb{C}$ to $\mathbb{C}$. Mapping something to a conjugate defines an isomorphism. This always happens.

An isomorphism $\phi : E \to E'$ *fixes* $a \in E$ if $\phi(a) = a$, and *fixes* a subfield F if it fixes all a in F. An isomorphism $E \to E'$ that fixes a subfield F is an *F-isomorphism.*

Theorem 48.2. *Let α be algebraic over F.*

i). For every conjugate β of α over F, there is a unique F-isomorphism $\Psi_{\alpha,\beta}$ of $F(\alpha)$ to $F(\beta)$.

ii). Every F-isomorphism of $F(\alpha)$ into $\overline{F}$ maps α to a conjugate β, so is $\Psi_{\alpha,\beta}$.

Proof. Let β be a conjugate of α over F, and let $f(x)$ be their common irreducible polynomial. The evaluation homomorphisms $\phi_\alpha : F[x] \to F(\alpha)$ and $\phi_\beta : F[x] \to F(\beta)$ have the same kernal $N = \langle f(x) \rangle$. The natural isomorphisms $\mu_\alpha : F(x)/N \to F(\alpha)$ and $\mu_\beta : F(x)/N \to F(\beta)$ fix F and map x to α and β respectivley, so the isomorphism $\Psi_{\alpha,\beta} := \mu_\beta \circ \mu_\alpha^{-1}$ is an F isomorphism of $F(\alpha)$ to $F(\beta)$. That this is unique is because an F-isomorphism on $F(\alpha)$ determined by its action on α.

Now let $\phi : F(\alpha)$ be an F-isomorphism of $F(\alpha)$ to a subfield of $\overline{F}$, and let $\beta = \phi(\alpha)$. Where $f(x) = \mathrm{irr}(\alpha, F)$ we have $f(\beta) = f(\phi(\alpha)) = \phi(f(\alpha)) = 0$ so β is conjugate to α, thus ϕ is $\Psi_{\alpha,\beta}$. □

As $a + ib$ and $a - ib$ are conjugate algebraic elements over $\mathbb{R}$, we have that they are roots of the same polynomials over $\mathbb{R}$. As $\sqrt{2}$ and $-\sqrt{2}$ have the same irreducible polynomial over $\mathbb{Q}$, they are conjugate over $\mathbb{Q}$. The $\mathbb{Q}$-map $\mathbb{Q}(\sqrt{2}) \to \mathbb{Q}(-\sqrt{2}) = \mathbb{Q}(\sqrt{2})$ taking $\sqrt{2}$ to $-\sqrt{2}$ is an isomorphism of $\mathbb{Q}(\sqrt{2})$ to itself.

An isomorphism of a field to itself is an *automorphism*, if it fixes a subfield F it is an F-automorphism. A set S of automorphisms of E *fixes* a subfield F if every $\sigma \in S$ fixes F.

Example 48.3. The $\mathbb{Q}$-automorphism $\Psi : \mathbb{Q}(\sqrt{2}, \sqrt{3})$ that takes $\sqrt{2}$ to $-\sqrt{2}$ and $\sqrt{3}$ to $\sqrt{3}$ fixes $\mathbb{Q}(\sqrt{3})$.

The automorphisms of a field tell us a lot about its subfields. Let $\mathrm{Aut}(E)$ be the set of automorphisms of a field E. For a subfield F of E let $\mathrm{Aut}(E/F) \subset \mathrm{Aut}(E)$ be the F-automorphisms of E. For $S \subset \mathrm{Aut}(E)$ let E_S be the set of elements of E fixed by S.

Theorem 48.4. *The set $\mathrm{Aut}(E)$ is a group under composition. The set $\mathrm{Aut}(E/F)$ is a subgroup of $\mathrm{Aut}(E)$. For $S \subset \mathrm{Aut}(E)$ the set E_S is a subfield of E.*

Proof. Three simple exercises. □

As it is a group, the text uses the notation $G(E/F)$ for $\mathrm{Aut}(E/F)$, and so I will try to too. The field E_S is the *fixed field* of S.

Example 48.5. Consider the extension $E = \mathbb{Q}(\sqrt{2}, \sqrt{3})$ of $\mathbb{Q}$. Any automorphism of E fixes $\mathbb{Q}$ as it must fix 1 and so the whole prime subfield of E. An automorphism is thus defined by its action on $\sqrt{2}$ and $\sqrt{3}$ as they generate E over Q, moreover an automorphism must either fix these degree two elements, or take them to their one conjugate, so there are 4 possible automorphisms: the identity , σ_2 taking $\sqrt{2}$ to $-\sqrt{2}$ and fixing $\sqrt{3}$, σ_3 taking $\sqrt{3}$ to $-\sqrt{3}$ and fixing $\sqrt{2}$, and $\sigma_6 = \sigma_2 \circ \sigma_3$. It is easy to check that all of these automorpisms other than have order 2 so $G(E/\mathbb{Q})$ is $\mathbb{Z}_2 \times \mathbb{Z}_2 = \langle \sigma_2, \sigma_3 \rangle$. The subgroup $\{\}$ fixes E, the subgroup $\langle \sigma_2 \rangle$ fixes $\mathbb{Q}(\sqrt{3})$. the subgroup$\langle \sigma_3 \rangle$ fixes $\mathbb{Q}(\sqrt{2})$ and the subgroup $\{\sigma_6\}$ fixes $\mathbb{Q}(\sqrt{6})$.

Problems from the Text

Section 48: 5, 7, 8, 11, 12, 16, 17, 20, 23, 24, 25, 29, 34, 35, 39

49 The Isomorphism Extension Theorem

For map $\tau : A' \to B'$ *extends* a map $\sigma : A \to B$ for some some $A \subset A'$ and $B \subset B'$ if $\tau(a) = \sigma(a)$ for all a in A.

Lemma 49.1. *For field isomorphism $\sigma : F \to F'$ and algebraic element α over F there is an extension of σ to an isomorphism τ of $F(\alpha)$ to a subfield of $\overline{F'}$.*

Proof. Extending σ to $\sigma' : F[x] \to F'[x]$ by letting it map x to x defines an isomorphism, and it is clear that σ' maps $f(x) = \mathrm{irr}(\alpha, F)$ to some irreducible polynomial $f'(x)$ in $F'[x]$. Let β be a zero of $f'(x)$. Then the evaluations maps $\phi_\alpha : F[x] \to F(\alpha)$ and $\phi'_\beta : F'[x] \to F'(\beta)$ have kernels $\langle f(x) \rangle$ and $\langle f'(x) \rangle$ respectively, and σ' induces an isomorphism between them so induces an isomorphism $\sigma' : F[x]/\langle f(x) \rangle \to F'[x]/\langle f'(x) \rangle$. Where $\mu : F[x]/\langle f(x) \rangle \to F(\alpha)$ and $\mu' : F'[x]/\langle f'(x) \rangle \to F(\beta)$ are the canonical isomorphisms, we have that $\mu' \circ \sigma \circ \mu^{-1}$ is an isomorphism of $F(\alpha)$ to $F(\beta)$. $\square$

Applying the above lemma iteratively we can easily show that any field isomorphism $F \to F'$ can be extended to take any finite extension E of F to $\overline{F'}$. When we go beyond finite extensions we have to worry about Zorn's Lemma type things again. We skip the proof of this, but it is in the text.

Theorem 49.2. *(Isomorphism Extension Theorem) Let E be an algebraic extension of a field F and let σ be an isomorphism of F to another field F'. Then σ extends to an isomorphism of E to a subfield of $\overline{F'}$.*

Let $\overline{F}$ and $\overline{F'}$ be two algebraic closures of a field F. Applying the Isomorphism Extension Theorem with $E = \overline{F}$ and σ the identity mapping of $F \to F$,

we get that σ extends to an F-isomorphism τ of $\overline{F}$ to a subfield of $\overline{F'}$. Applying it then to τ^{-1} extends this to an isomorphism of $\overline{F'}$ to $\overline{F}$. But this is already onto so τ was actually isomorphism. So we get the following.

Corollary 49.3. *For a field F the algebraic closure is unique up to F-isomorphism.*

Now really, any algebraic extension E of F can be viewed as a subfield of the unique algebraic closure $\overline{F}$.

The index of a field extension

For algebraic extension E of F, the *index* $\{E : F\}$ *of* E *over* F is the number of F-isomorphisms of E to subfields of $\overline{F}$. The isomorphism extension theorem extends each of these to F-isomorphisms of $\overline{F}$ into $\overline{F}$, so F-automorphisms of $\overline{F}$.

Fact 49.4. *For $E \in \overline{F}$, the index $\{E : F\}$ is the number different mappings of E under F-automorphisms of $\overline{F}$.*

If $F \leq E \leq K$ are finite extensions, each of the $\{E : F\}$ F-isomorphisms τ of E into $\overline{F}$ extend to $\{K : E\}$ different extensions of τ to F-isomorphism of K into $\overline{F}$, so we have $\{K : E\}\{E : F\}$ different F-isomorphism of K. Thus the following is immediate from Theorem 49.2.

Corollary 49.5. *If $F \leq E \leq K$ are finite extensions then*

$$\{K : F\} = \{K : E\}\{E : F\}.$$

For finite extension $E \geq F$ we have

$$|G(E/F)| \leq \{E : F\} \leq [E : F].$$

Indeed $\{E : F\}$ counts the images of E under F-automorphisms of $\overline{F}$, but these are only counted by $|G(E/F)|$ if they map E onto E. On the other hand, there is only one F-isomorphism of $E = F(\alpha)$ for each distinct root of $\mathrm{irr}(\alpha, F)$ so there are at most $[F(\alpha) : F]$, giving that $\{E : F\} \leq [E : F]$ for simple extensions. Using the mutliplicativity of dimensions and indices, the inequality holds for all finite extensions.

Example 49.6. Where $E = \mathbb{Q}(\sqrt{2}, \sqrt{3})$ we showed in Example 31.4 that the degree $[E : \mathbb{Q}]$ is 4 by showing that $\{1, \sqrt{2}, \sqrt{3}, \sqrt{6}\}$ was a basis of E over $\mathbb{Q}$. So we know that $\{E : \mathbb{Q}\}$ is at most 4. In Example 48.5 we showed that there are four $\mathbb{Q}$-automorphisms of E, so $\{E : \mathbb{Q}\}$ is at least 4. Thus $\{E : \mathbb{Q}\} = 4$. In fact, we showed that $\{E : \mathbb{Q}(\sqrt{2})\} = 2$ and $\{\mathbb{Q}(\sqrt{2}) : \mathbb{Q}\} = 2$, which is an example of Corollary 49.5.

Problems from the Text

Section 49: 1-3,4,5,6,7,8

50 Splitting Fields

We saw that for finite extensions E of F the following hold

$$|G(E/F)| \leq \{E : F\} \leq [E : F];$$

we look know at when we have $|G(E/F)| = \{E : F\}$.

A polynomial $f(x) \in F[x]$ *splits* in an extension E of F if all of its zeros are in E. For a set $P = \{f_i(x) \mid i \in I\}$ of polynomials in $F[x]$, a field $E \leq \overline{F}$ is the *splitting field for P over F* if it is the smallest extension of F in which all polynomials in P split. If $P = \{f(x)\}$, we say that E *is splitting for $f(x)$*. Note that the splitting field of a set $P \subset F[x]$ is indeed unique, as it is the intersection of all fields in $\overline{F}$ that split all of the polynomials. A field $E \leq \overline{F}$ is a splitting field over F if it is the splitting field for some set $P \subset F[x]$.

Example 50.1. The field $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ is splitting over $\mathbb{Q}$ as it is splitting for $\{x^2 - 2, x^2 - 3\}$. It is also splitting for $\{x^4 - 5x^2 + 6\}$.

Theorem 50.2. *An extension $E \leq \overline{F}$ of F is splitting over F if and only if every F-automorphism of $\overline{F}$ restricts on E to an automorphism of E.*

Proof. Assume that E is splitting for a set $P \subset F[x]$. As any F-automorphism ϕ of $\overline{F}$ must map zeros of $f(x) \in P$ to zeros of $f(x)$, any F-automorphism maps the set Z of all zeros of all polynomials in P to Z. So ϕ is an F-automorphism of $F(Z) \subset E$. As $F(Z)$ contains all zeros of all polynomials in P, it is splitting for P, so is E.

On the other hand, assume that every F-automorphism of $\overline{F}$ maps E onto E. For any element $\alpha \in E$, and any other zero β of $\mathrm{irr}(\alpha, F)$ in $\overline{F}$ there is an F-isomorphism of E taking α to β. This extends, by the isomorphism extension theorem, to an isomorphism τ of $\overline{F}$ to $\overline{F}$. So τ is an F-automorphism of $\overline{F}$, so maps E to E by assumption. Thus all roots of $\mathrm{irr}(\alpha, F)$ are in E. So E splits the set $P = \{\mathrm{irr}(\alpha, F) \mid \alpha \in E\}$. As every element of E is a root of some $f(x)$ in P, E is splitting for P. $\square$

We will use this last part of the proof again. Let's state it as a corollary.

Corollary 50.3. *If $E \leq \overline{F}$ is a splitting field over F then it splits* $\mathrm{irr}(F, \alpha)$ *for all $\alpha \in E$.*

Recall that $\{E : F\}$ is the number different mappings of E under F-automorphisms of $\overline{F}$. We have just showed that if E is splitting then all of these are F-automorphisms of E, so the following is immediate.

Corollary 50.4. *For finite extension $E \geq F$,*

$$|G(E/F)| = \{E : F\},$$

if and only if E is splitting over F.

The following example is from Example 50.9 of the text, but also includes several of the exercises from that chapter.

Example 50.5. The polynomial $f(x) = x^3 - 2$ is irreducible over $\mathbb{Q}$ and has root $\gamma_0 := \sqrt[3]{2}$ but does not split over $\mathbb{Q}(\sqrt[3]{2})$ as its other roots

$$\gamma_1 = \sqrt[3]{2}\left(\tfrac{-1+i\sqrt{3}}{2}\right) \qquad \text{and} \qquad \gamma_2 = \sqrt[3]{2}\left(\tfrac{-1-i\sqrt{3}}{2}\right)$$

are not even in $\mathbb{R} \supset \mathbb{Q}(\sqrt[3]{2})$. So $\mathbb{Q}(\sqrt[3]{2})$ is not splitting over $\mathbb{Q}$. Verifying Corollary 50.4 we see that

$$|G(\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q})| = 1 < 3 = \{\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}\}.$$

Indeed, any $\mathbb{Q}$-automorphism of $\mathbb{Q}(\sqrt[3]{2})$ is determined by which of the γ_i it takes γ_0 to, but only γ_0 is in $\mathbb{Q}(\sqrt[3]{2})$, giving the first equality. For any γ_i we get a different $\mathbb{Q}$-isomorphism of $\mathbb{Q}(\sqrt[3]{2})$, giving the second.

Now $f(x)$ does split in $E := \mathbb{Q}(\sqrt[3]{2}, i\sqrt{3})$. As the splitting field for $f(x)$ must contain $\sqrt[3]{2}$ and γ_1, and a field with these must contain $i\sqrt{3}$, we get that E is splitting for $f(x)$. We show that $6 = |G(E/\mathbb{Q})|$ and $[E : \mathbb{Q}] = 6$ so by

$$6 = |G(E/\mathbb{Q})| \leq \{E : \mathbb{Q}\} \leq [E : \mathbb{Q}] = 6$$

get that $\{E : \mathbb{Q}\} = 6$ and that $|G(E/\mathbb{Q})| = \{E : \mathbb{Q}\}$ as fortold by Corollary 50.4.

Indeed, it is easy to check that $\{1, i\sqrt{3}\}$ is a basis of E over $\mathbb{Q}(\sqrt[3]{2})$ and so

$$[E : \mathbb{Q}] = [E : \mathbb{Q}(\sqrt[3]{2})] \cdot [\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 2 \cdot 3 = 6;$$

thus $\{E : \mathbb{Q}\} \leq 6$. On the other hand, clearly $|G(\mathbb{Q}(i\sqrt{2})/\mathbb{Q})| = 2$, and $|G(\mathbb{Q}(\sqrt[3]{2}, i\sqrt{2})/\mathbb{Q}(i\sqrt{2}))| = 3$ as for every zero γ_i of $f(x)$ there is a $\mathbb{Q}(i\sqrt{2})$-automorphism of $\overline{F}$ taking γ_0 to γ_i, and all of these are in $\mathbb{Q}(\sqrt[3]{2}, i\sqrt{2})$ so induce an automorphism. So

$$|G(E/\mathbb{Q})| = |G(E/\mathbb{Q}(i\sqrt{3}))| \cdot |G(\mathbb{Q}(i\sqrt{3})/\mathbb{Q})| = 3 \cdot 2.$$

Problems from the Text

Section 50: 5, 6, 10, 11-14, 16, 18, 21, 22

51 Separable Extensions

We consider the problem of when, for extension E of F,

$$\{E : F\} = [E : F].$$

The main ideas are seen by considering $E = F(\alpha)$ where $f(x) = \mathrm{irr}(\alpha, F)$ has degree n. In this case the dimension $[E : F]$ is n, and the action of any F-automorphism of $\overline{F}$ on E is determined by is action on α. It must take α to a zero of $f(x)$ so $\{E : F\}$ is the number of different zeros of $f(x)$. So we get $\{E : F\} = [E : F]$ if and only if the n zeros of $f(x)$ are distinct.

Multiplicity of zeros of a polynomial

A zero $\alpha \in \overline{F}$ of $f(x) \in F[x]$ has *multiplicity* ν if ν is the largest integer for which $(x - \alpha)^\nu$ divides $f(x)$ in $\overline{F}$.

As for two zeros of irreducible $f(x) \in F[x]$ there is an F-automorphism of $\overline{F}$ taking α to β, $(x - \alpha)^d$ divides $f(x)$ if and only if $(x - \beta)^d$ does.

Fact 51.1. *All zeros in $\overline{F}$ of an irreducible polynomial in $F[x]$ have the same multiplicity. So any irreducible monic polynomial in $F[x]$ can be written*

$$f(x) = \prod_i (x - \alpha_i)^\nu$$

for some $\nu \geq 1$ where α_i runs over the distinct zeros of $f(x)$.

Applying this to $f(x) = \mathrm{irr}(\alpha, F)$ for some $\alpha \in \overline{F}$, we get that the degree of $f(x)$ is the number n of distinct zeros of $f(x)$ times their multiplicity ν. From the discussion above we get that

$$[E : F] = \{E : F\}\nu.$$

As any finite extension E of F can be written as $E = F(\alpha_1, \ldots, \alpha_n)$ for finite n, we get, where $E_i = F(\alpha_1, \ldots, \alpha_i)$, that

$$[E : F] = [E_n : E_{n-1}] \ldots [E_1 : F] = \{E_n : E_{n-1}\}\nu_n \ldots \{E_1 : F\}\nu_1 = \{E; F\} \prod \nu_i.$$

Theorem 51.2. *For any finite extension E of F, $\{E : F\}$ divides $[E : F]$.*

The following shows that there are irreducible polynomials with zeros of multiplicity greater than 1.

Example 51.3. Observe that where $F = \mathbb{Z}_p(t)$ for an indeterminant t, the polynomial $x^p - t \in F[x]$ and any root y has multiplicity p. Indeed, if y is a root then $y^p = t$ and so $x^p - t = x^p - y^p = (x - y)^p$ showing that y has multiplicity p. That $x^p - t$ is irreducible, comes from the fact that if it is not, it has factor $(x - y)^d$ for some d with $1 \leq d < p$. But expanding this will yield some monomials in x with co-efficients y^i for $i = 1, \ldots, d$, and these are not in F.

Separable Extensions

A finite extension E of F is *separable over* F if $[E : F] = \{E, F\}$. Element $\alpha \in \overline{F}$ is *separable* over F if $F(\alpha)$ is separable over F. An irreducible polynomial $f(x) \in F[x]$ is *separable over* F if every zero of $f(x)$ is separable over F.

As $[F(\alpha) : F]$ is the degree of $\mathrm{irr}(\alpha, F)$ and $\{F(\alpha) : F\}$ is then number of distinct zeros of $\mathrm{irr}(\alpha, F)$, we have the following.

Fact 51.4. *A irreducible polynomial $f(x)$ in $F[x]$ is separable over F if and only if all zeros have multliplicity* 1.

Example 51.5. The polynomial $f(x) = x^3 - 2$ which is irreducible over $\mathbb{Q}$ has three distinct roots in $\overline{\mathbb{Q}}$, so $\mathbb{Q}(\sqrt[3]{2})$ is separable. Indeed we previously argued $\{\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}\} = 3$ and $[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 3$. We also saw that $|G(\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q})| = 1$ as the other roots are not in $\mathbb{Q}(\sqrt[3]{2})$. So this is not a splitting extension.

We can break the problem of deciding if a finite extension is separable to simple extensions.

Lemma 51.6. *If $F \leq E \leq K$ are finite extensions, then K is separable over F if and only if K is separable over E and E is separable over F.*

Proof. This is trivial as

$$[K : F] = [K : E][E : F] \qquad \text{and} \qquad \{K : F\} = \{K : E\}\{E : F\}.$$

□

Problem 51.1. Trivial Pursuit: Really prove the above lemma.

Theorem 51.7. *For finite $E \geq F$, E is separable over F if and only if each $\alpha \in E$ is separable over F.*

Proof. If E is separable over F and $\alpha \in E$, then $E \geq F(\alpha) \geq F$ and so α is separable by the previous lemma. On the other hand, if each $\alpha \in E$ is separable over F, then as we can build E as $F(\alpha_1, \alpha_2, \ldots, \alpha_d)$ by a series of simple extensions, by the previous lemma it is enough to observe that α_i is separable over $F' = F(\alpha_1, \alpha_2, \ldots, \alpha_{i-1})$. But this is clear as $f'(x) = \mathrm{irr}(\alpha_i, F')$ divides $f(x) = \mathrm{irr}(\alpha_i, F)$ so α_i has multiplicity in $f'(x)$ because it has multiplicity 1 in $f(x)$. □

Perfect Fields and the primitive element theorem

A field F is *perfect* if every finite extension is separable.

In this section we show the following two theorem.

Theorem 51.8. *If F has characteristic* 0*, or is finite, then it is perfect.*

Theorem 51.9. *(The Primitive Element Theorem) Every finite separable extension of a field is simple.*

The main idea of the proof is to show that if $F(\alpha, \alpha')$ is generated by a single element β. We use the fact that their irreducible polynomials have distinct roots.

But we skip this so we can get to the punch.

Problems from the Text

Section 51: 2, 4, 7, 8, 9, 11, 12, 13

53 Galois Theory

Recall for a finite extension $K \geq F$ we have

$$|G(K/F)| \leq \{K : F\} \leq [K : F].$$

Inside $\overline{F}$, every F-isomorpism of E must permute the roots of $\text{irr}(\alpha, F)$ for any α in E, but these roots might not all be in E. If the are in E for every $\alpha \in E$, then K is splitting, and we have equality on the left. If the roots are distinct for each α, then K is separable, and we have equality on the right.

A finite extension $K \geq F$ is *finite normal* if it is splitting and separable.

Theorem 53.1. *If $K \geq E \geq F$ where K is a finite normal extension of F, then*

i). K is a finite normal extension of E,

ii). $G(K/E)$ is the subgroup of $G(K/F)$ of elements fixing E, and

iii). two elements of $G(K/F)$ restrict to the same isomorphism of E if and only if there are in the same left coset of $G(K/E)$ in $G(K/F)$.

Proof. For i), if K is splitting for some $P \subset F[x]$ it is splitting for this same P viewed as polynomials in $E[x]$. If K is separable over F then it is separable over E by Lemma 51.6.

Certainly any element of $G(K/E)$ fixes F, so is an element of $G(K/F)$ fixing E, so ii) is trivial. Now take two elements σ and τ of $G(K/F)$. They restrict to the same isomorphism of E if and only if $\tau^{-1} \circ \sigma$ fixes E, so if and only if $\tau^{-1} \circ \sigma \in G(K/E)$. But this holds if and only if $\tau G(K/E) = \sigma G(K/E)$. $\square$

Now talking about cosets $G(K/E)$ is teasing us with its normality. Then they go build the tension by calling a field extension 'normal'. They have me on the edge of my seat!

Galois' theorem says that if $K \geq F$ is normal, then there is a one-to-one correspondence $E \mapsto G(K/E)$ between the subextensions E, and the subgroups of $G(K/F)$ that fix them. It takes normal subextensions to normal subgroups. Lets see an example before we state it in detail.

Example 53.2. The extension $K = \mathbb{Q}(\sqrt{2}, \sqrt{3})$ of $\mathbb{Q}$ is normal. As $\{1, \sqrt{2}, \sqrt{3}, \sqrt{6}\}$ is a basis of K over $\mathbb{Q}$, the automorphism group $G(K/Q)$ is $\{, \sigma_2, \sigma_3, \sigma_6 = \sigma_2 \circ \sigma_3\}$ where σ_2 and σ_3 fix $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\sqrt{3})$ respectively. The subgroups $G(K/Q), \{, \sigma_2\}, \{, \sigma_3\}, \{, \sigma_6\}$ and $\{\}$ respectively fix the subextensions $\mathbb{Q}$, $\mathbb{Q}(\sqrt{2}), \mathbb{Q}(\sqrt{3}), \mathbb{Q}(\sqrt{6})$ and K. The group $G(K/Q) \cong \mathbb{Z}_2 \times \mathbb{Z}_2$ is abelian, so all subgroups are normal. Thus all the subextensions are normal extensions of $\mathbb{Q}$.

If K is a finite normal extension of F then $G(K/F)$ is the *Galois group of K over F*.

Theorem 53.3. *Let K be a finite normal extension of F and for a subextension E let $\lambda(E) = G(K/E)$ be the subgroup of $G = G(K/F)$ fixing E. The map λ is a one-to-one correspondence between the fields E with $F \leq E \leq K$ and the subgroups of G. Moreover:*

i). For subextension E, $E = K_{\lambda(E)}$

ii). For subgroup $H \leq G$, $\lambda(K_H) = H$.

iii). $[K : E] = |\lambda(E)|$ and $[E : F] = (G(K/F) : \lambda(E))$.

iv). E is a normal extension of F if and only if $\lambda(E)$ is a normal subgroup of G. In this case $G(E/F) \cong G(K/F)/(G(K/E))$.

Proof. For part i) we have clearly that $E \leq K_{\lambda(E)}$. Any element $\alpha \in K - E$, splits over F so over E. So there is an E-automorphism of $\overline{F}$ that takes it to a conjugate. Thus α is not in $K_{\lambda(E)}$.

For part ii) we clearly have that H is contained in $\lambda(K_H) = G(K/K_H)$. We show the containment is exact. As $K \geq F$ is normal so is $K \geq K_H$

$$n := [K : K_H] = \{K : K_H\} = |G(K : K_H)| \geq |H|$$

Towards contradiction, assume that $|H| < n$. As E is finite normal, so separable, K_H is finite separable over K, so is simple: $K_H = K(\alpha)$. Consider the polynomial

$$f(x) \in \prod_{\sigma \in H} (x - \sigma(\alpha)).$$

As $\sigma H = H$ for each $\sigma \in H$, the coefficients of each monomial in $f(x)$ are invariant under the action of σ, so $f(x)$ is in $K_H[x]$. Further $(\alpha) = \alpha$ for the identity element $\in H$, so $f(\alpha) = 0$, showing that $\mathrm{irr}(\alpha, K_H)$ divides $f(x)$, and so

$$\deg(\alpha, K_H) < n = [K : K_H] = [K_H(\alpha) : K_H],$$

which is a contradiction.

For part iii), we have from Theorem 53.1 that $K \geq E$ is normal because $K \geq F$ is so $|\lambda(E)| = |G(K/E)| = [K : E]$. That $[E : F] = (G(K/F) : \lambda(E))$ is then immediate from multiplicativity of dimensions and the normality of $K \geq F$:

$$(G(K/F) : \lambda(E))|\lambda(E)| = |G(K/F)| = [K : F] = [K : E][E : F].$$

For part iv), we must show that $G(K/E)$ is normal in $G(K/F)$ if and only if E is normal. As $K > F$ is normal, so separable, $E > F$ is separable, so we have to show that $G(K/E)$ is normal if and only if E is splitting, that is if and only if, for all α in E and all $\sigma \in G(K/F)$, that $\sigma(\alpha)$ is in E.

Well $G(K/E)$ is normal in $G(K/F)$ if and only if for all $\sigma \in G(K/F)$ and $\tau \in G(K/E)$ we have $\sigma^{-1}\tau\sigma \in G(K/E)$. This means exactly that for any $\alpha \in E$, we have $\sigma^{-1}\tau\sigma(\alpha) = \alpha$, ie. that $\tau\sigma(\alpha) = \sigma(\alpha)$. But this holds for all $\tau \in G(K/E)$ if and only if $\sigma(\alpha)$ is in the fixed field of $G(K/E)$, which is E. $\square$

Galois groups over finite fields

Let $F = \mathrm{GF}(p^r)$, and let K be a finite (so separable) extension of dimension $[K : F] = n$. Then $K = \mathrm{GF}(p^{rn})$ and is splitting for $x^{p^{rn}} - x$, so is normal. The automorphism

$$\sigma_{p^r} : K \to K : \alpha \mapsto \alpha^{p^r}$$

leaves F fixed by Euler's little theorem. If $\sigma_{p^r}^d$ fixes all elements of K then they are all roots of $x^{p^{rd}} - x$. but this can have at most p^{rd} roots, so σ_{p^r} has order at least n. As $|G(K/F)| = [K : F] = n$ it follows that σ_{p^r} generates $G(K/F)$.

Theorem 53.4. *If K is a finite extension of $\mathrm{GF}(p^r)$ of dimension n then $G(K/F)$ is cyclic of order n and generated by $\sigma_{p^r} : \alpha \mapsto \alpha^{p^r}$.*

Example 53.5. Let $F = \mathbb{Z}_p$ and $K = \mathrm{GF}(p^{12})$. Then $G(K/F) \cong (\mathbb{Z}_{12}, +)$ is generated by σ_p. The subgroups are $\langle \sigma_p^d \rangle$ for all d dividing 12. Observe that for d dividing p the map σ_p^d, which takes α to α^{p^d}. The elements of K fixed by this are the roots of $x^{p^d} - x$ so are the elements of $\mathrm{GF}(p^d)$.

Problems from the Text

Section 53:

54 Illustrations of Galois Theory

We look at several finite normal field extensions, and reveal the lattice of subextensions by finding their Galois groups, and finding the fixed fields of their

subgroups.

$\mathbb{Q}(i, \sqrt[4]{2})$ over $\mathbb{Q}$

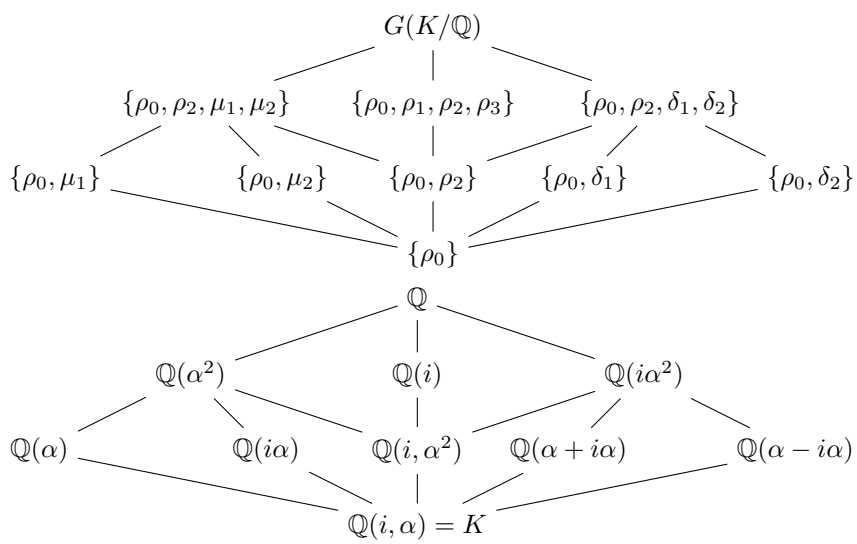
Consider the irreducible polynomial $f(x) = x^4 - 2$ in $\mathbb{Q}[x]$. It has roots $\{\pm\alpha, \pm i\alpha\}$ where $\alpha = \sqrt[4]{2}$. The splitting field K of $f(x)$ contains i and α so is $K = \mathbb{Q}(\alpha, i)$. (It clearly cannot be smaller.) It is a normal extension as it is a finite extension of the perfect field $\mathbb{Q}$. It has dimension

$$[\mathbb{Q}(\alpha, i) : \mathbb{Q}] = [\mathbb{Q}(\alpha, i) : \mathbb{Q}(\alpha)] \cdot [\mathbb{Q}(\alpha) : \mathbb{Q}] = 2 \cdot 4 = 8.$$

Its basis over $\mathbb{Q}$ is $\{1, i\} \times \{\alpha, \alpha^2, \alpha^3, \alpha^4\}$ and any σ in $G(K/\mathbb{Q})$ is determined by its action on these basis elements, which are determined by which of their conjugates it takes i and α to: $\sigma(i) \in \{\pm i\}$ and $\sigma(\alpha) \in \{\pm\alpha, \pm i\alpha\}$. From this one can show that $G(K/\mathbb{Q})$ is the dihedral group with table

σ	ρ_0	ρ_1	ρ_2	ρ_3	μ_1	δ_1	μ_2	δ_2
$\sigma(\alpha)$	α	$i\alpha$	$-\alpha$	$-i\alpha$	α	$i\alpha$	$-\alpha$	$-i\alpha$
$\sigma(i)$	i	i	i	i	$-i$	$-i$	$-i$	$-i$

Compare the group diagram of $G = G(K/\mathbb{Q})$ and the field diagram of the corresponding subextensions of $K > \mathbb{Q}$ that they fix.



Showing which fields are fixed by which groups can take some work. We show a couple of them.

To find the field K_1 fixed by $H_1 = \{\rho_0, \rho_1, \rho_2, \rho_3\}$ we have to find an extension of degree $(G : H) = 2$ that is fixed by all of the ρ_i. Clearly $\mathbb{Q}(i)$ is this field. To

find the field K_2 fixed by $H_2 = \{\rho_0, \mu_1\}$ we have to find an extension of degree 4 that is fixed under $i \mapsto -i$. Well this fixes α so $\mathbb{Q}(\alpha)$ and this has the right dimension, so it is the fixed field.

Finding the dimension 4 extension field K_3 fixed by $H_3 = \{\rho_0, \delta_1\}$ is harder. As H_3 is a group , it fixes $\rho_0(\beta) + \delta_1(\beta)$ for any $\beta \in K$. Trying various β we find that $\rho_0(\alpha) + \delta_1(\alpha) = \alpha + i\alpha =: \gamma$ is fixed. So $Q(\gamma)$ is fixed by H_3. To verify that it has dimension 4, we can check that it has four conjugates by acting the other automorphisms on it: $i\alpha - \alpha, -\alpha - i\alpha$, and $i\alpha + \alpha$. This also allows us to find $\mathrm{irr}(\gamma, \mathbb{Q})$.

$\mathbb{Q}((1+i)/\sqrt{2})$ over $\mathbb{Q}$

The polynomial $f(x) = x^4 + 1$ has roots $(\pm 1 \pm i)/\sqrt{2}$. A bit of knowledge of complex numbers allows us to write these as $\alpha, \alpha^3, \alpha^5$, and α^7 where $\alpha = (1+i)/\sqrt{2}$, so it is clear that all are in $K = \mathbb{Q}(\alpha)$, and that K is the splitting field of $f(x)$. So $K > \mathbb{Q}$ is normal. The $\mathbb{Q}$-automorphisms of K are determined by their action on α, so are $G = G(K/\mathbb{Q}) = \{\sigma_1, \sigma_3, \sigma_5, \sigma_7\}$ where $\sigma_i(\alpha) = \alpha^i$. All elements of G have order 2, so it is $\mathbb{Z}_2 \times \mathbb{Z}_2$.

Compare again the diagrams for G and sub-extensions of $K > \mathbb{Q}$.

$$\begin{tikzcd}
 & G(K/\mathbb{Q}) \arrow[dl, dash] \arrow[d, dash] \arrow[dr, dash] & \\
\{\sigma_1, \sigma_3\} \arrow[dr, dash] & \{\sigma_1, \sigma_5\} \arrow[d, dash] & \{\sigma_1, \sigma_7\} \arrow[dl, dash] \\
 & \{\sigma_1\} &
\end{tikzcd}
\qquad
\begin{tikzcd}
 & \mathbb{Q} \arrow[dl, dash] \arrow[d, dash] \arrow[dr, dash] & \\
\mathbb{Q}(i\sqrt{2}) \arrow[dr, dash] & \mathbb{Q}(i) \arrow[d, dash] & \mathbb{Q}(\sqrt{2}) \arrow[dl, dash] \\
 & \mathbb{Q}(\alpha) &
\end{tikzcd}$$

All but the trivial extensions have dimension 2. Again to see that $\{\sigma_1, \sigma_3\}$ fixes $i\sqrt{2}$ we observe that $\sigma_1(\alpha) + \sigma_2(\alpha) = \alpha + \alpha^3 = i\sqrt{2}$. That $\{\sigma_1, \sigma_7\}$ fixes $\sqrt{2}$ is similar. To see that $\{\sigma_1, \sigma_5\}$ fixes i observe that $\alpha\alpha^5 = -i$.

Symmetric Functions

As any group is a subgroup of a symmetric group, the following shows how any group is the Galois Group of some field.

In the field $F^{(n)} = F(y_1, \ldots, y_n)$, view S_n as the group of permutations of the indeterminants. Any permutation is an F-automorphism, and those elements that are fixed by all permutations are F-automorphisms called the *symmetric functions* in $y_1, \ldots, y_n$ over F. Let $K = F^{(n)}_{S_n}$ be the field consisting of these functions. The polynomial $f(x) = \prod_{i=1}^n (x - y_i)$ of degree n in $F^{(n)}[x]$ has co-efficients in K, so is in $K[x]$. The coefficients are called the *elementary symmetric functions* in the y_i. The first is $s_1 = \sum_{i=1}^n y_i$, the second is $s_2 =$

$\sum_{(i,j)\in\binom{n}{2}} y_i y_j$, etc. Let $E = F(s_1, s_2, \ldots, s_n)$. Clearly $E \leq K \leq F^{(n)}$, and indeed $F^{(n)}$ is a finite normal extension of E, being the splitting field of $f(x)$. As $f(x)$ has degree n we have $[K : E] \leq n!$. On the other hand since $K = F^{(n)}_{S_n}$, $G(F^{(n)}/K) = S_n$ so

$$n! = |S_n| \leq [F^{(n)} : K] \leq [F^{(n)} : E] \leq n!.$$

Thus $K = E$, and S_n is $G(F^{(n)}/E)$.

This shows the following.

- Every symmetric function in $y_1, \ldots, y_n$ over F is a rational function in the elementary symmetric functions s_i.
- The Galois group of $F(s_1, \ldots, s_n)$ is S_n.

Problems from the Text

Section 54: